US 20250282665A1

(19) United States

(12) Patent Application Publication

MAEHARA et al.

(10) Pub. No.: US 2025/0282665 A1

(43) Pub. Date: Sep. 11, 2025

(54) **GLASS MANUFACTURING APPARATUS AND GLASS MANUFACTURING METHOD**

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(21) Appl. No.: **19/217,678**

(22) Filed: **May 23, 2025**

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2023/040616, filed on Nov. 10, 2023.

(30) **Foreign Application Priority Data**

Nov. 25, 2022 (JP) 2022-188270

Publication Classification

(51) **Int. Cl.**
C03B 5/237 (2006.01)
C03B 5/235 (2006.01)

(52) **U.S. Cl.**
CPC ***C03B 5/237*** (2013.01); ***C03B 5/2353*** (2013.01)

(57) **ABSTRACT**

A glass manufacturing apparatus (**1**) includes a recuperator (**60**) configured to heat a heat medium through heat exchange with a combustion gas emitted from a glass melting furnace (**10**), a first reactor (**71**) configured to produce CH_4 gas and H_2O gas from CO_2 gas and H_2 gas by performing a methanation reaction, an water vapor controller (**73**) configured to adjust an H_2O concentration in a product of the first reactor by removing at least a part of the H_2O gas from the product of the first reactor, a second reactor (**72**) configured to produce CO gas and H_2 gas from the product of the first reactor with the H_2O concentration adjusted in the water vapor controller by performing at least one of a dry reforming reaction and a steam reforming reaction through heat exchange with the heat medium heated by the recuperator, and a burner (**20**) configured to form a flame in the glass melting furnace by burning a oxidizing gas and a flammable gas containing the CO gas and the H_2 gas produced by the second reactor.

1

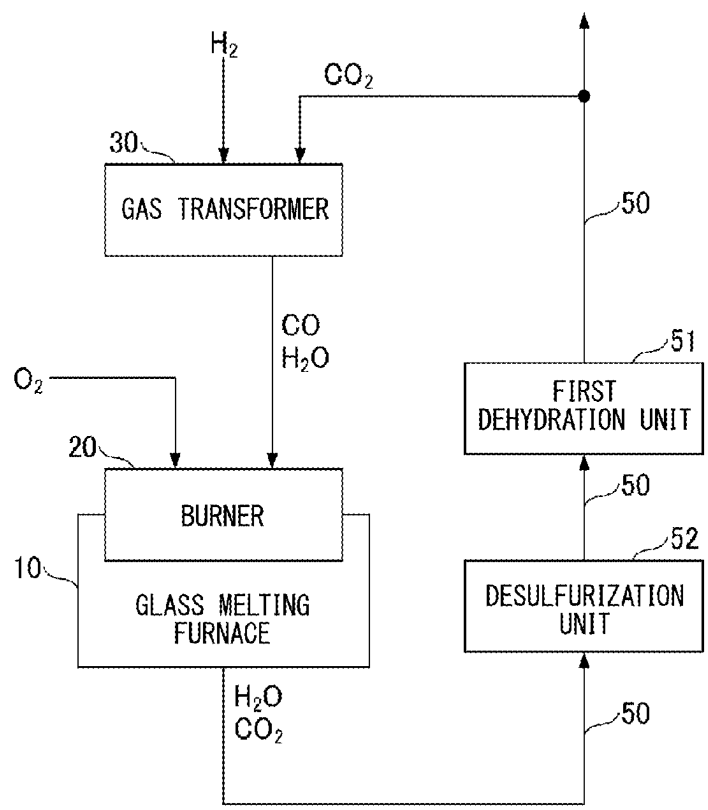


FIG. 1

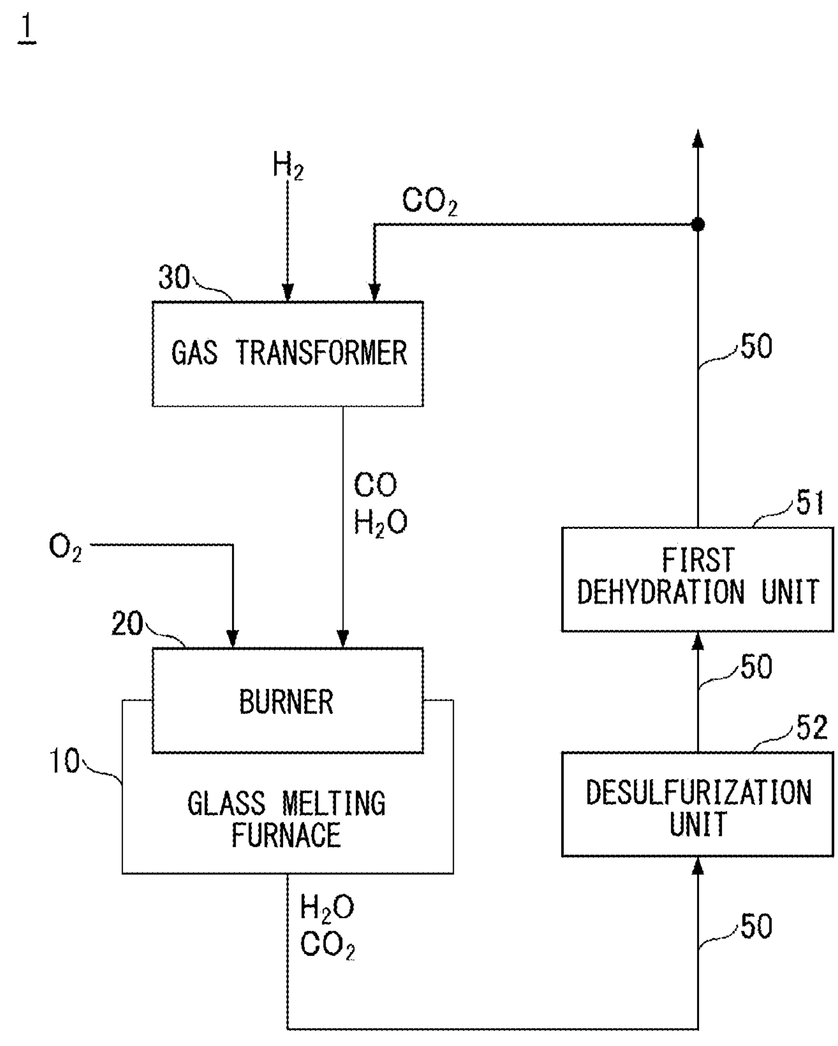
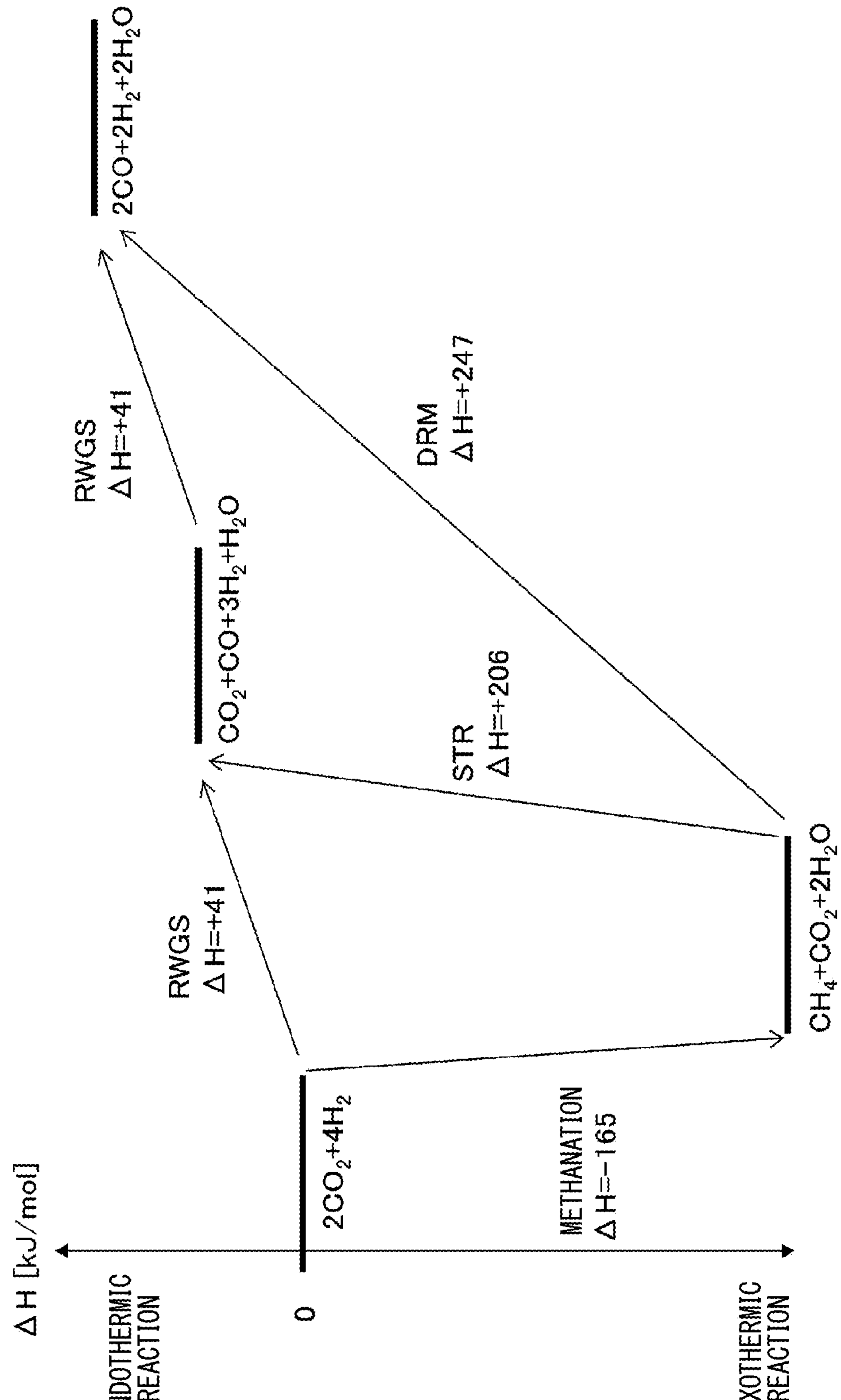

1
H2
CO2
30
GAS TRANSFORMER
50
CO
H2O
O2
51
FIRST DEHYDRATION UNIT
20
BURNER
50
52
10
GLASS MELTING FURNACE
DESULFURIZATION UNIT
H2O
CO2
50

FIG. 2



ΔH [kJ/mol]
ENDOTHERMIC REACTION
0
EXOTHERMIC REACTION
$2CO_2+4H_2$
METHANATION
ΔH=−165
$CH_4+CO_2+2H_2O$
RWGS
ΔH=+41
STR
ΔH=+206
$CO_2+CO+3H_2+H_2O$
RWGS
ΔH=+41
DRM
ΔH=+247
$2CO+2H_2+2H_2O$

FIG. 3

	EXAMPLE 1	EXAMPLE 2	EXAMPLE 3	EXAMPLE 4	EXAMPLE 5	EXAMPLE 6	EXAMPLE 7	EXAMPLE 8	EXAMPLE 9	EXAMPLE 10	EXAMPLE 11	EXAMPLE 12	EXAMPLE 13	EXAMPLE 14
FLAMMABLE GAS	CH_4	CH_4	H_2	H_2	H_2	H_2	H_2	H_2	H_2	H_2	H_2	H_2	H_2	H_2
PRESENCE/ABSENCE OF TRANSFORMATION OF FLAMMABLE GAS	PRESENCE	PRESENCE	PRESENCE	PRESENCE	ABSENCE	ABSENCE	ABSENCE	ABSENCE	ABSENCE	ABSENCE	ABSENCE	ABSENCE	ABSENCE	ABSENCE
MOLAR RATIO x	–	–	–	–	0.00	0.25	0.50	0.55	0.75	1.00	2.00	3.00	4.00	5.00
OXIDIZING GAS	Air	O_2	Air	O_2	O_2	O_2	O_2	O_2	O_2	O_2	O_2	O_2	O_2	O_2
H_2O GAS CONCENTRATION [% BY VOLUME] IN FLAMMABLE GAS	19	67	35	100	100	80	67	65	57	50	33	25	20	17
H_2O GAS CONCENTRATION [% BY VOLUME] IN FURNACE ATMOSPHERE	20	61	35	88	88	73	62	61	55	49	34	26	21	18

FIG. 4

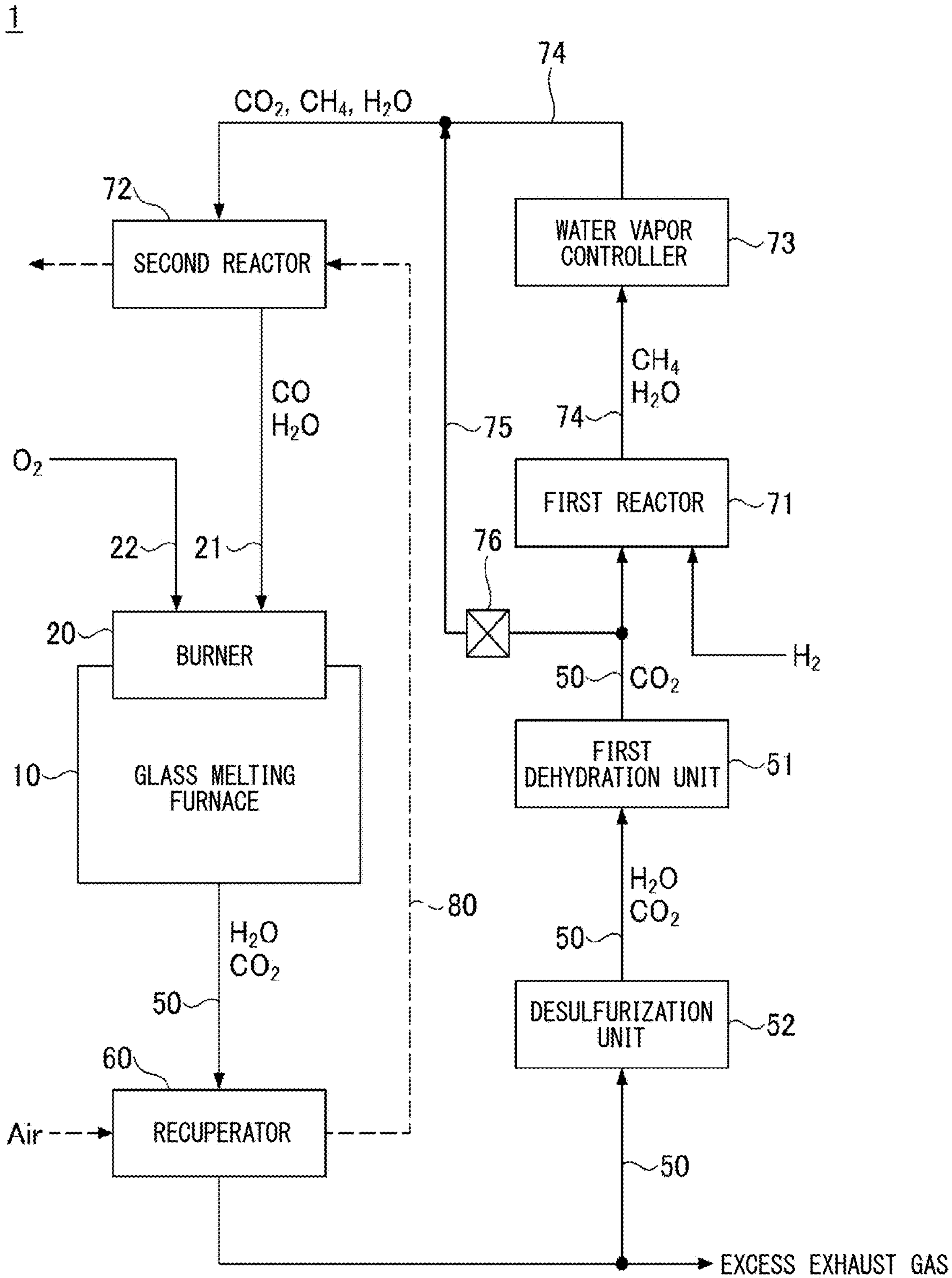

1
74
CO2, CH4, H2O
72
SECOND REACTOR
WATER VAPOR CONTROLLER
73
CO
H2O
CH4
H2O
74
75
O2
FIRST REACTOR
71
22
21
76
20
BURNER
50
CO2
H2
10
GLASS MELTING FURNACE
FIRST DEHYDRATION UNIT
51
H2O
CO2
80
H2O
CO2
50
50
DESULFURIZATION UNIT
52
60
Air
RECUPERATOR
50
EXCESS EXHAUST GAS

FIG. 5

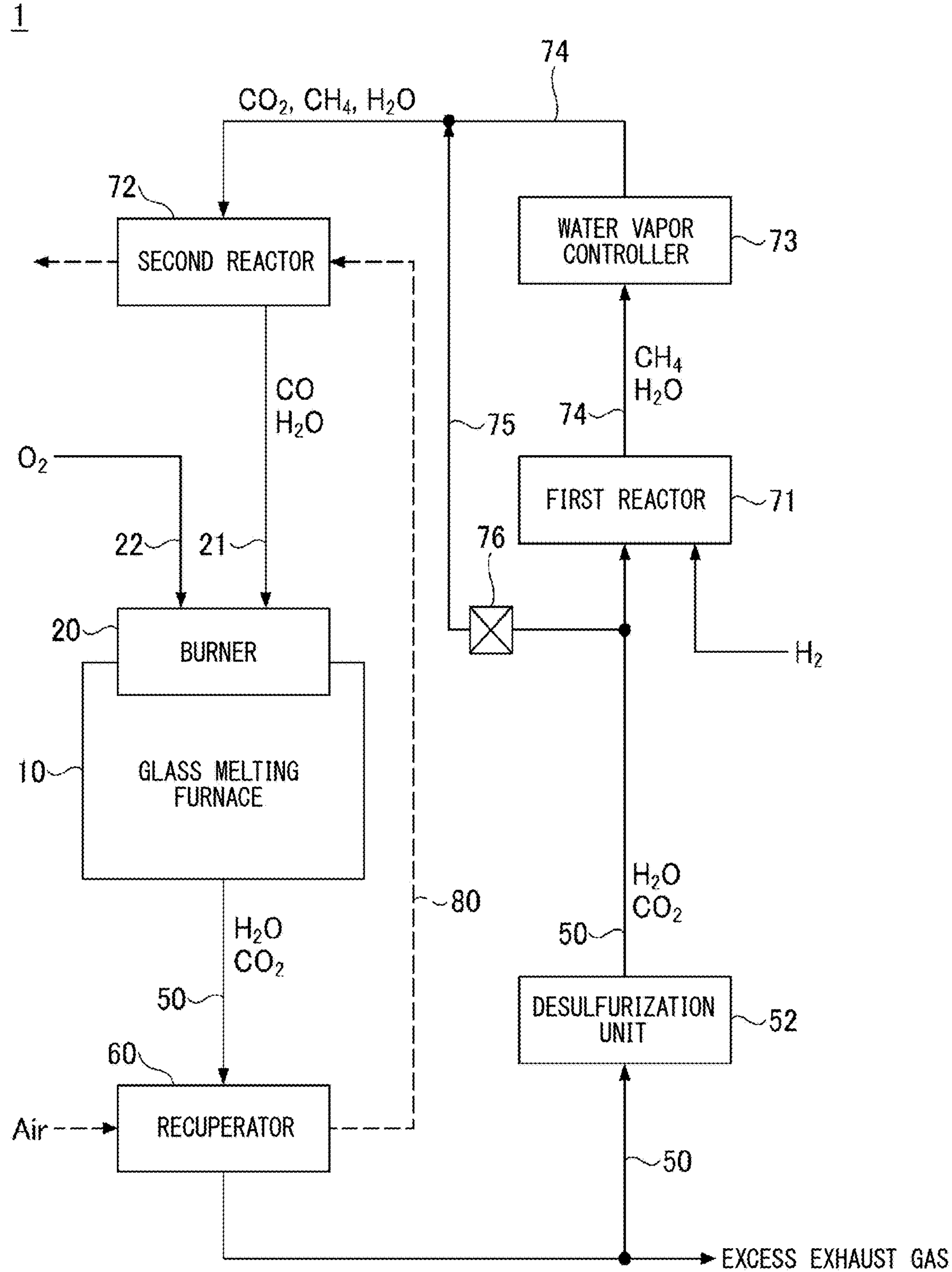

1
74
CO_2, CH_4, H_2O
72
SECOND REACTOR
WATER VAPOR CONTROLLER
73
CO
H_2O
CH_4
H_2O
75
74
O_2
FIRST REACTOR
71
22
21
76
20
BURNER
H_2
10
GLASS MELTING FURNACE
H_2O
CO_2
80
50
H_2O
CO_2
50
DESULFURIZATION UNIT
52
60
Air
RECUPERATOR
50
EXCESS EXHAUST GAS

FIG. 6

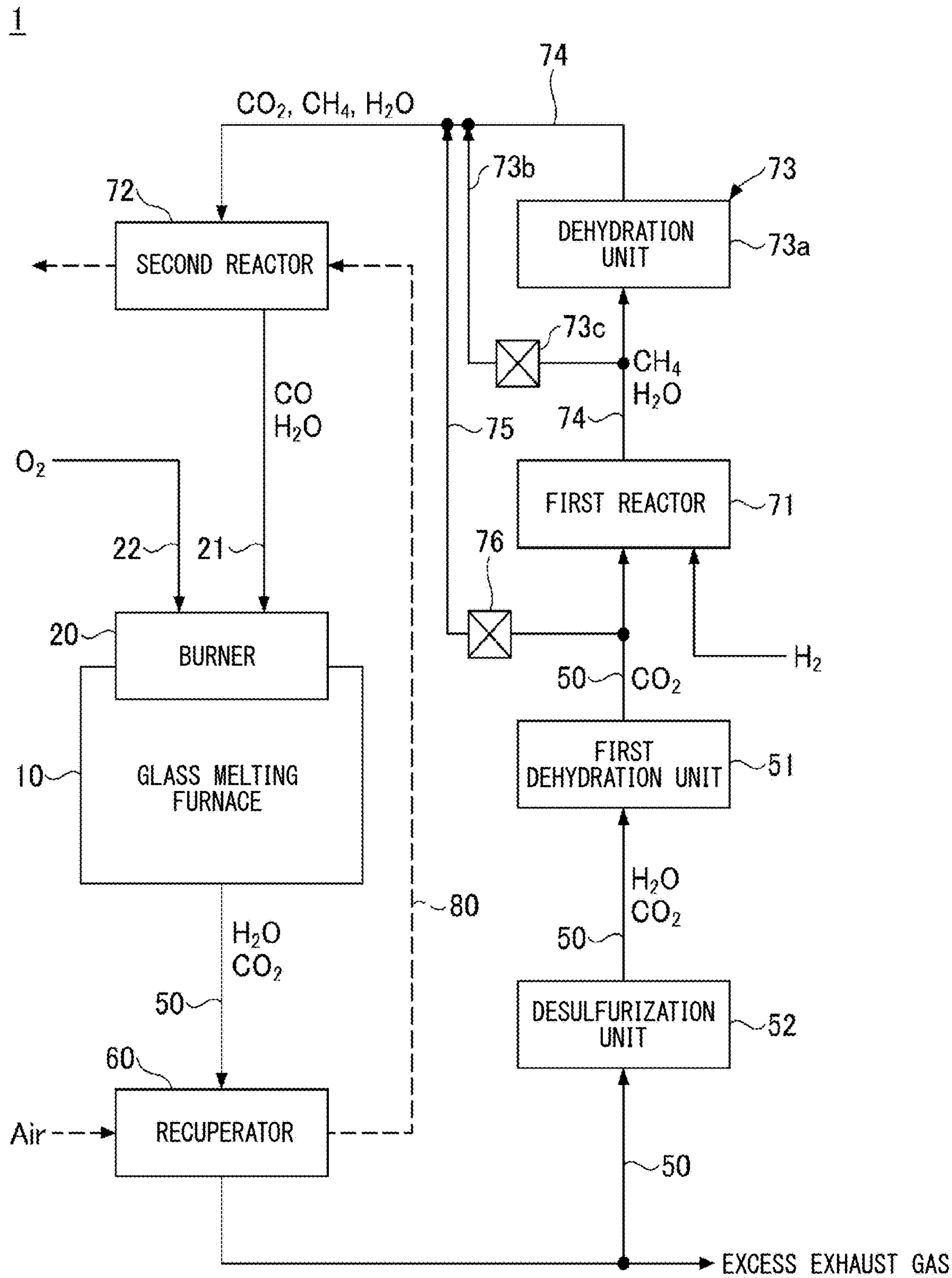
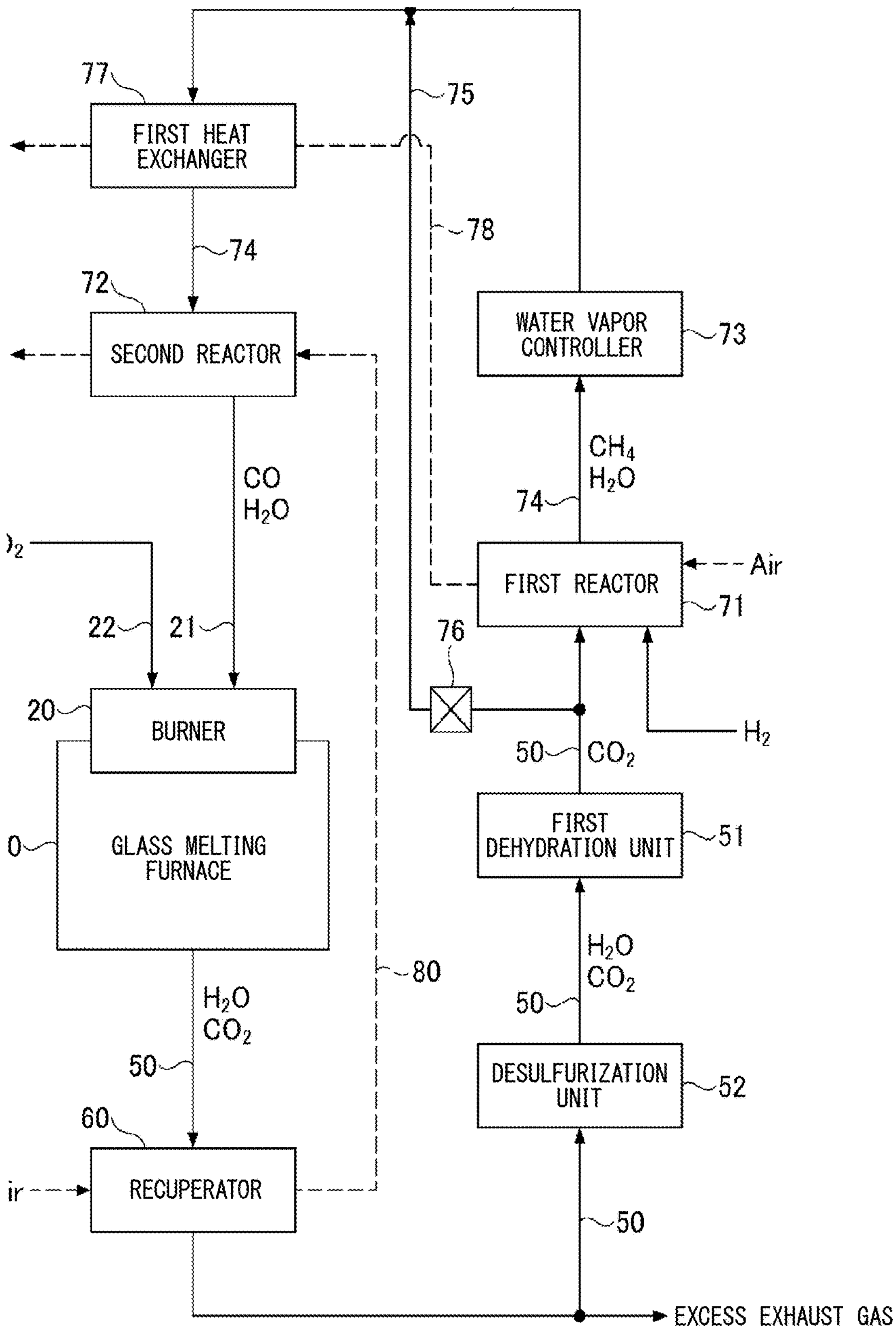

1
74
CO2, CH4, H2O
73b
73
72
DEHYDRATION UNIT
73a
SECOND REACTOR
73c
CH4
H2O
CO
H2O
75
74
O2
FIRST REACTOR
71
22
21
76
20
BURNER
H2
50
CO2
10
GLASS MELTING FURNACE
FIRST DEHYDRATION UNIT
51
80
H2O
CO2
H2O
CO2
50
50
DESULFURIZATION UNIT
52
60
Air
RECUPERATOR
50
EXCESS EXHAUST GAS

FIG. 7



77
FIRST HEAT EXCHANGER
75
78
74
72
SECOND REACTOR
WATER VAPOR CONTROLLER
73
CO
H_2O
CH_4
H_2O
74
)2
22
21
FIRST REACTOR
Air
71
76
20
BURNER
H_2
50
CO_2
0
GLASS MELTING FURNACE
FIRST DEHYDRATION UNIT
51
H_2O
CO_2
50
80
H_2O
CO_2
50
DESULFURIZATION UNIT
52
60
ir
RECUPERATOR
50
EXCESS EXHAUST GAS

FIG. 8

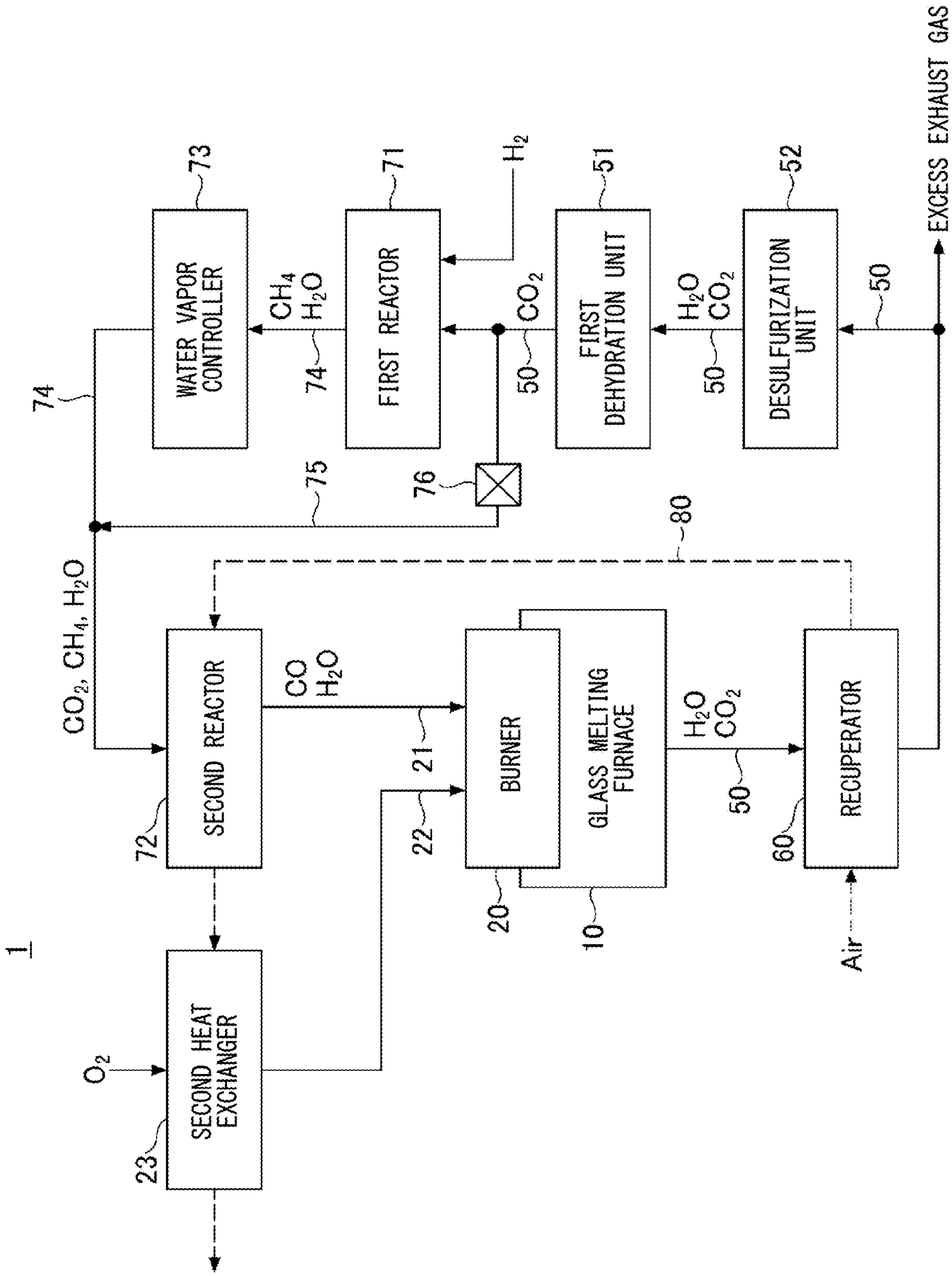
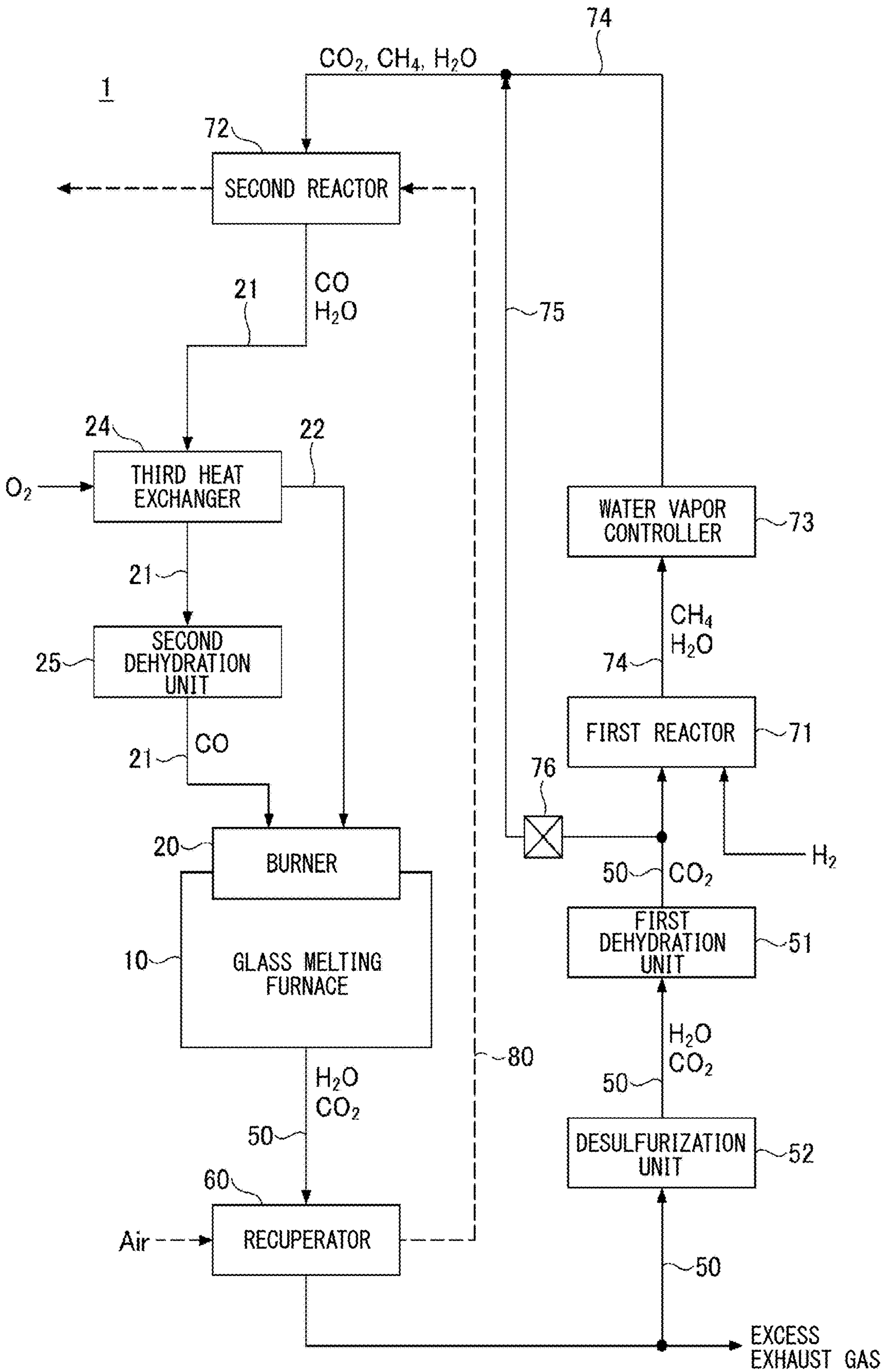

1
WATER VAPOR CONTROLLER
73
FIRST REACTOR
71
H_2
FIRST DEHYDRATION UNIT
51
DESULFURIZATION UNIT
52
EXCESS EXHAUST GAS
CH_4 H_2O
74
CO_2
50
H_2O CO_2
50
50
74
75
76
80
CO_2, CH_4, H_2O
SECOND REACTOR
72
CO H_2O
21
BURNER
20
GLASS MELTING FURNACE
10
H_2O CO_2
50
RECUPERATOR
60
Air
22
O_2
SECOND HEAT EXCHANGER
23

FIG. 9



1
74
CO_2, CH_4, H_2O
72
SECOND REACTOR
CO
H_2O
21
75
24
22
O_2
THIRD HEAT EXCHANGER
WATER VAPOR CONTROLLER
73
21
CH_4
H_2O
25
SECOND DEHYDRATION UNIT
74
FIRST REACTOR
71
21
CO
76
20
BURNER
H_2
50
CO_2
10
GLASS MELTING FURNACE
FIRST DEHYDRATION UNIT
51
H_2O
CO_2
80
50
H_2O
CO_2
50
60
DESULFURIZATION UNIT
52
Air
RECUPERATOR
50
EXCESS EXHAUST GAS

FIG. 10

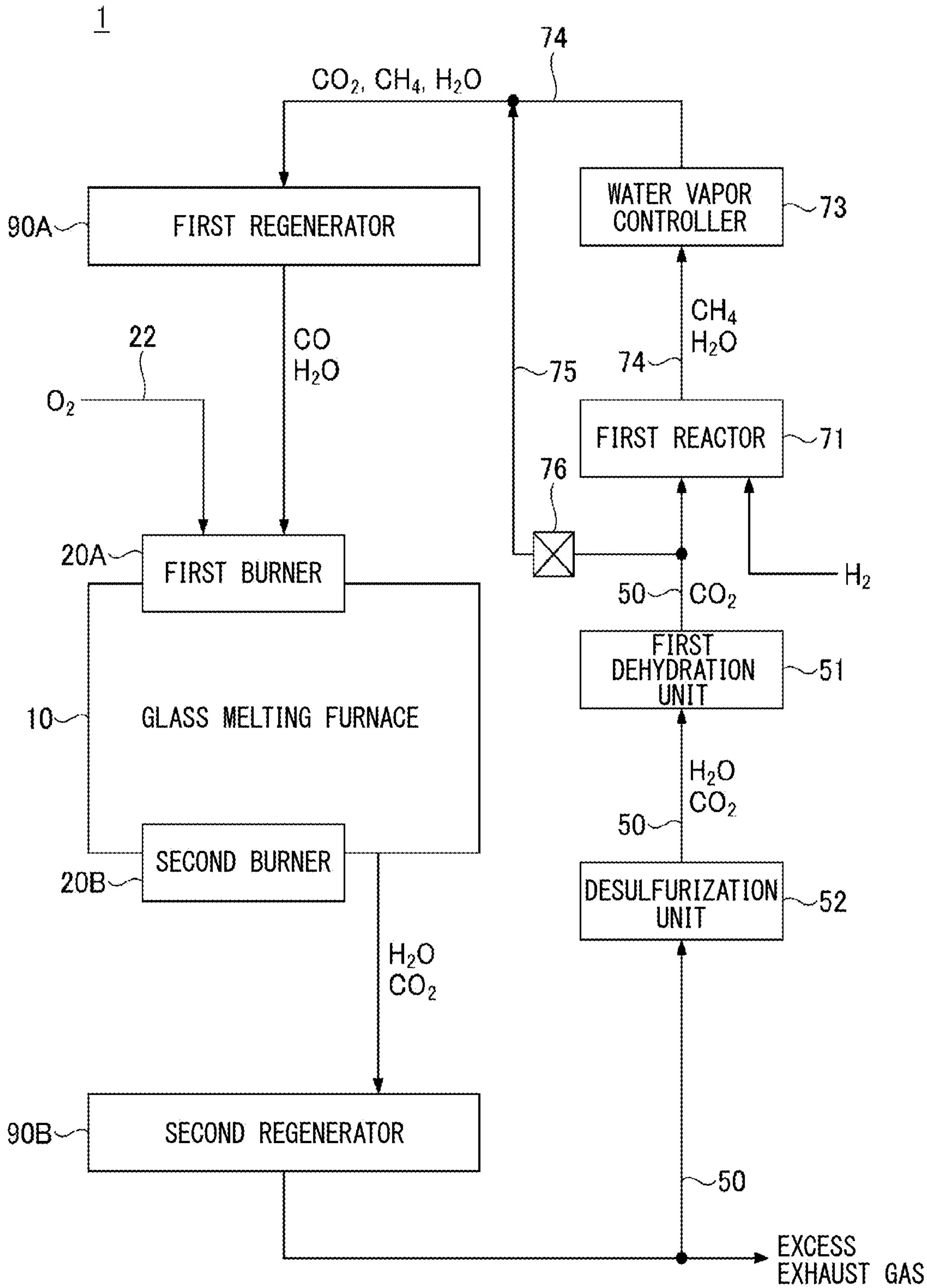

1
74
CO2, CH4, H2O
WATER VAPOR CONTROLLER
73
90A
FIRST REGENERATOR
CH4
H2O
22
CO
H2O
74
75
O2
FIRST REACTOR
71
76
20A
FIRST BURNER
H2
50
CO2
FIRST DEHYDRATION UNIT
51
10
GLASS MELTING FURNACE
H2O
CO2
50
SECOND BURNER
DESULFURIZATION UNIT
52
20B
H2O
CO2
90B
SECOND REGENERATOR
50
EXCESS EXHAUST GAS

FIG. 11

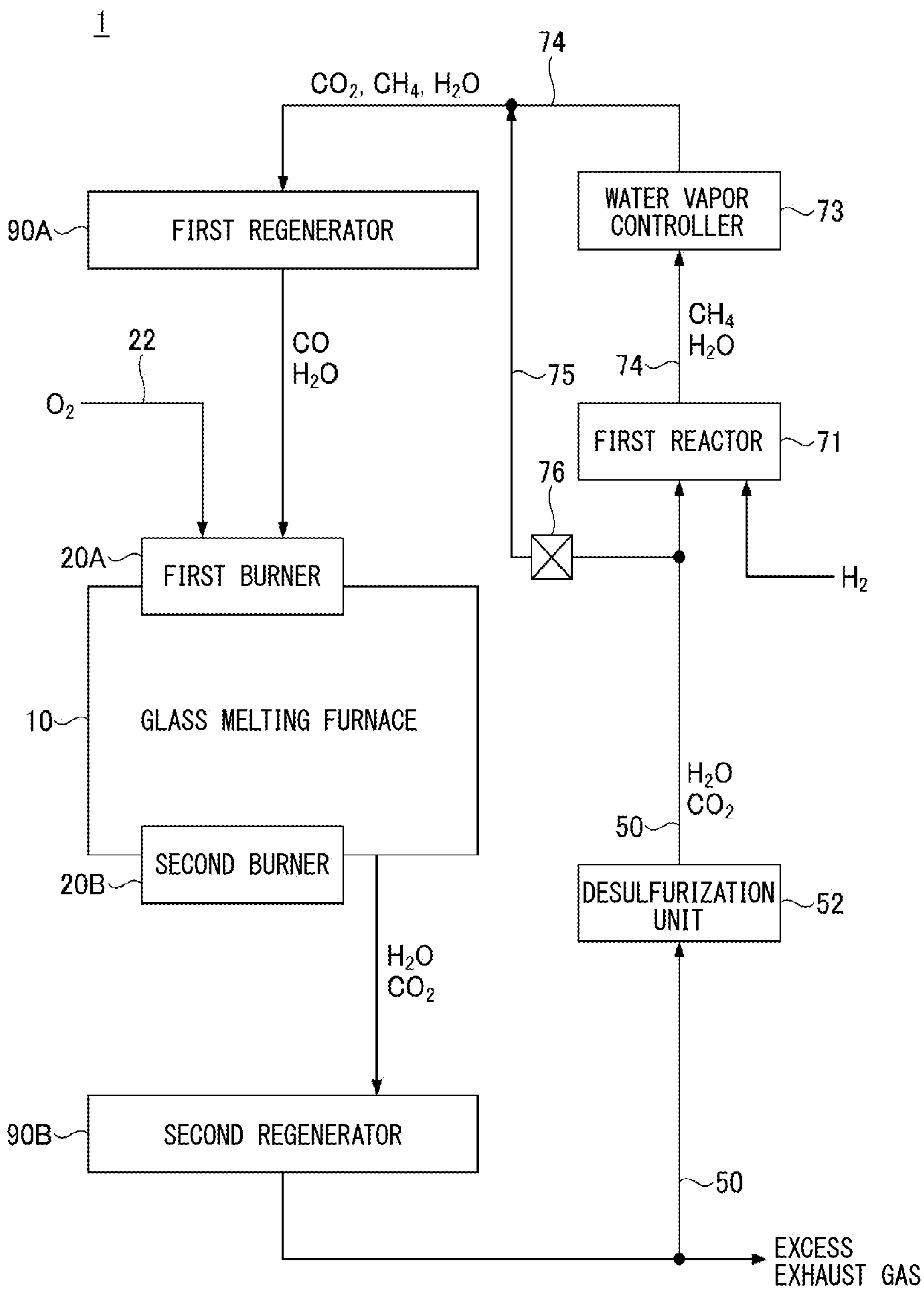

1
74
CO_2, CH_4, H_2O
90A
FIRST REGENERATOR
WATER VAPOR CONTROLLER
73
CO
H_2O
CH_4
H_2O
22
O_2
75
74
FIRST REACTOR
71
76
20A
FIRST BURNER
H_2
10
GLASS MELTING FURNACE
H_2O
CO_2
50
SECOND BURNER
20B
DESULFURIZATION UNIT
52
H_2O
CO_2
90B
SECOND REGENERATOR
50
EXCESS EXHAUST GAS

FIG. 12

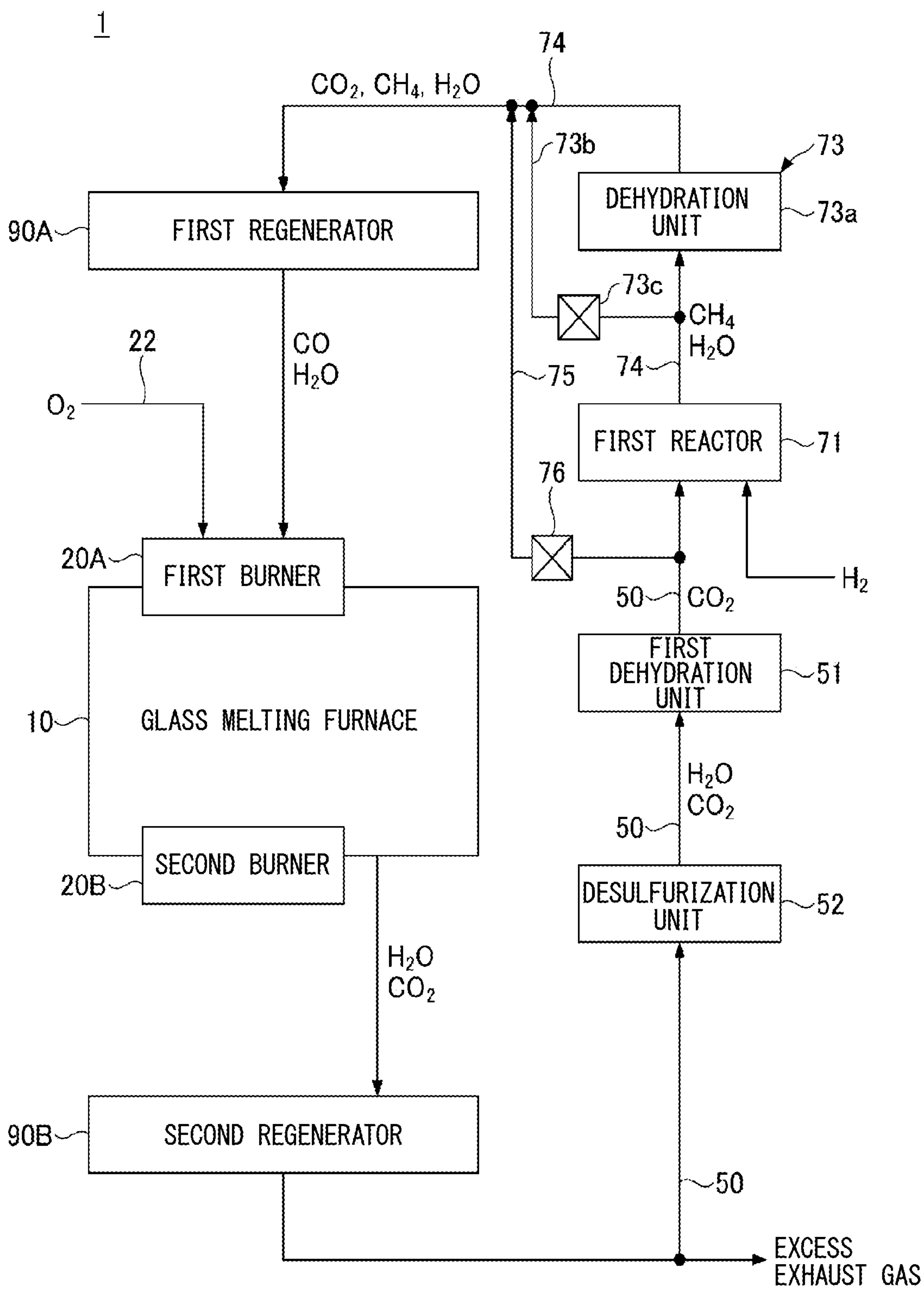

1
74
CO_2, CH_4, H_2O
73b
73
DEHYDRATION UNIT
73a
90A
FIRST REGENERATOR
73c
CH_4
H_2O
22
CO
H_2O
74
75
O_2
FIRST REACTOR
71
76
20A
FIRST BURNER
H_2
50
CO_2
FIRST DEHYDRATION UNIT
51
10
GLASS MELTING FURNACE
H_2O
CO_2
50
SECOND BURNER
20B
DESULFURIZATION UNIT
52
H_2O
CO_2
90B
SECOND REGENERATOR
50
EXCESS EXHAUST GAS

FIG. 13

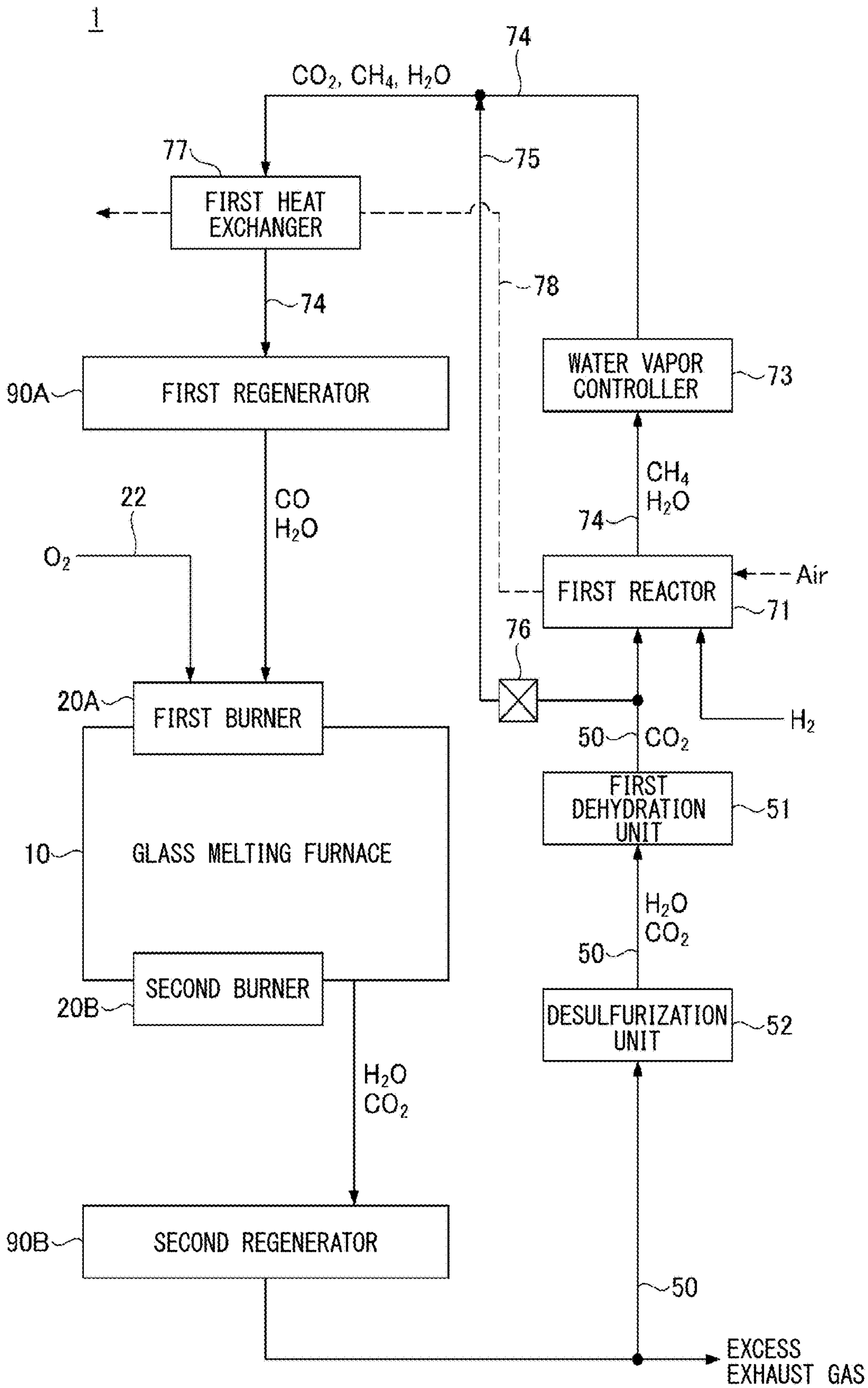

1
74
CO_2, CH_4, H_2O
77
75
FIRST HEAT EXCHANGER
78
74
90A
FIRST REGENERATOR
WATER VAPOR CONTROLLER
73
CO
H_2O
CH_4
H_2O
22
74
O_2
FIRST REACTOR
Air
71
76
20A
FIRST BURNER
H_2
50
CO_2
FIRST DEHYDRATION UNIT
51
10
GLASS MELTING FURNACE
H_2O
CO_2
50
SECOND BURNER
20B
DESULFURIZATION UNIT
52
H_2O
CO_2
90B
SECOND REGENERATOR
50
EXCESS EXHAUST GAS

FIG. 14

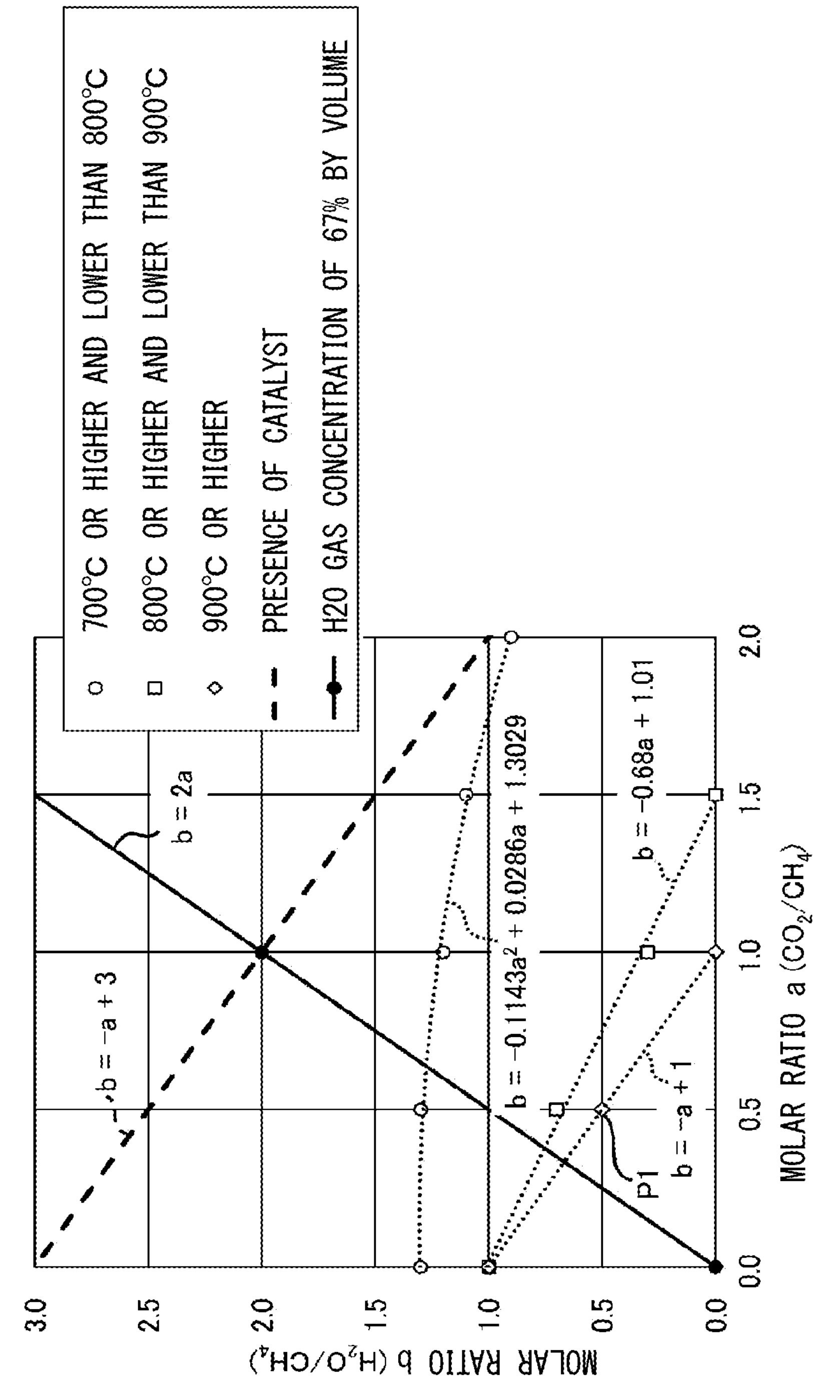

700°C OR HIGHER AND LOWER THAN 800°C
800°C OR HIGHER AND LOWER THAN 900°C
900°C OR HIGHER
PRESENCE OF CATALYST
H20 GAS CONCENTRATION OF 67% BY VOLUME
b = 2a
b = −a + 3
b = −0.1143a² + 0.0286a + 1.3029
b = −0.68a + 1.01
b = −a + 1
P1
MOLAR RATIO a (CO2/CH4)
MOLAR RATIO b (H2O/CH4)
0.0
0.5
1.0
1.5
2.0
2.5
3.0

FIG. 15

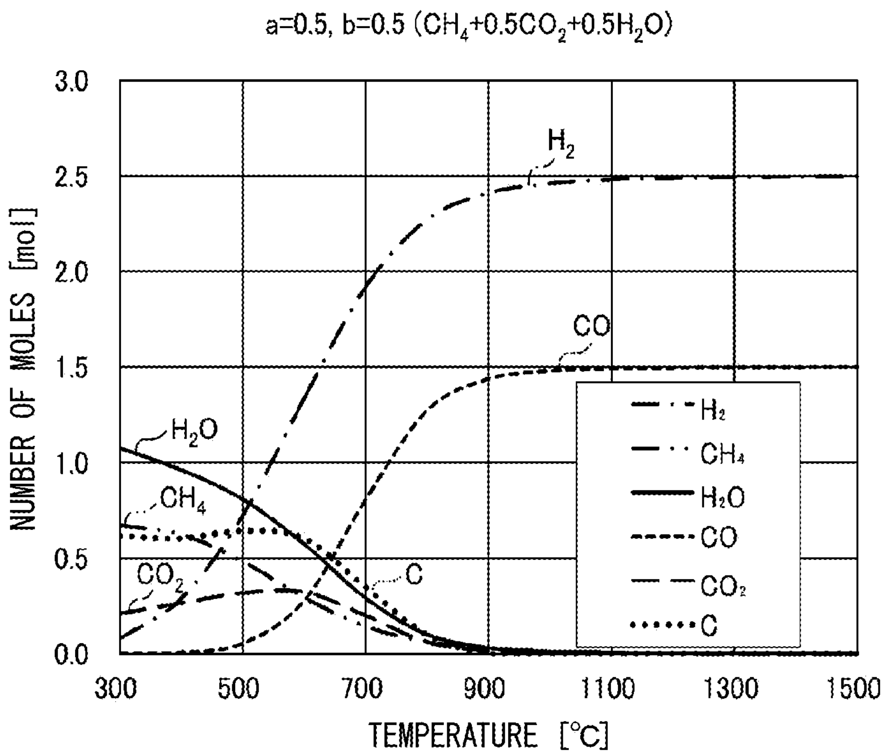
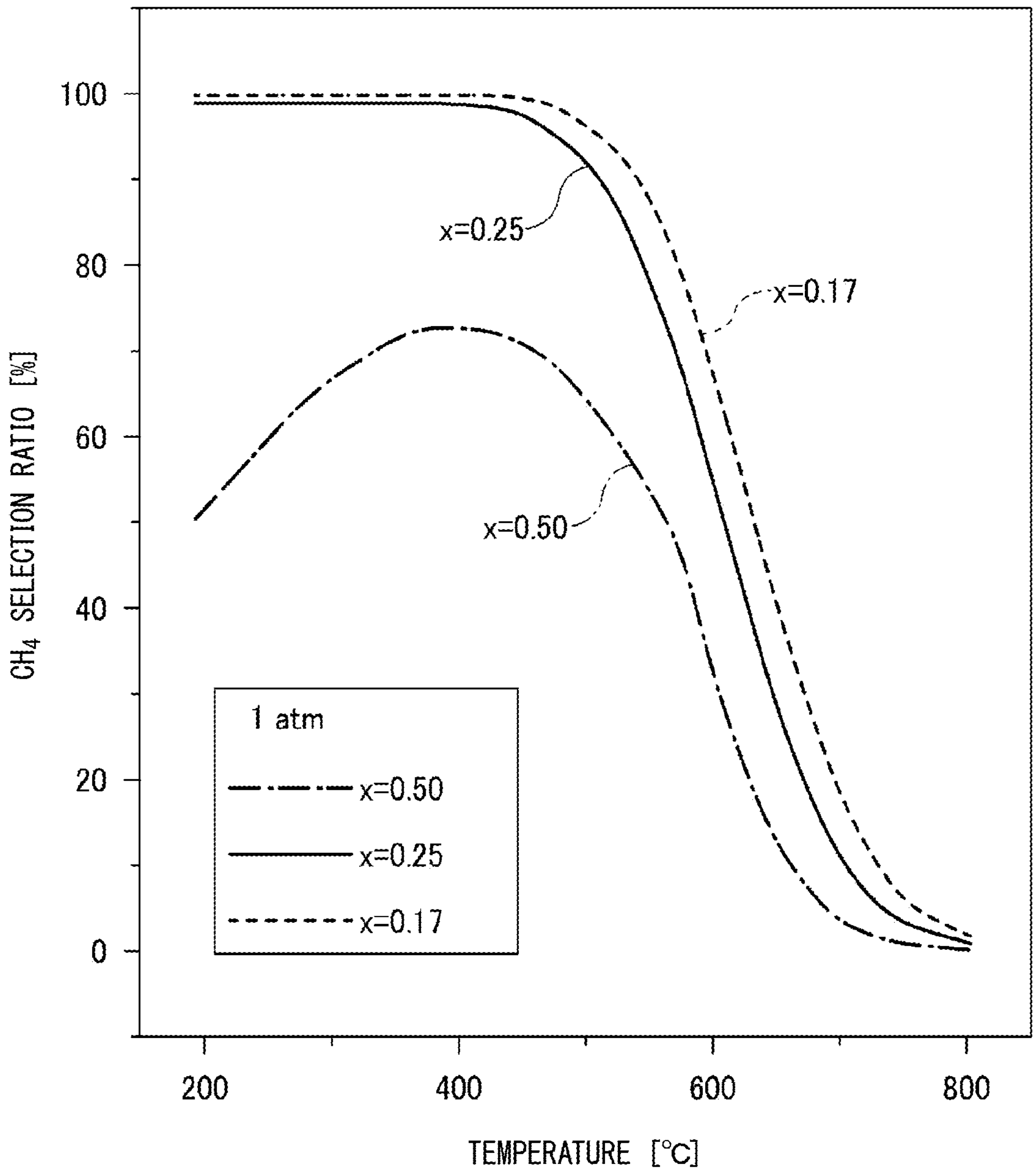

a=0.5, b=0.5 (CH4+0.5CO2+0.5H2O)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
NUMBER OF MOLES [mol]
H2
CO
H2O
CH4
CO2
C
H2
CH4
H2O
CO
CO2
C
300
500
700
900
1100
1300
1500
TEMPERATURE [°C]

FIG. 16


100
80
60
40
20
0
CH4 SELECTION RATIO [%]
200
400
600
800
TEMPERATURE [°C]
x=0.25
x=0.17
x=0.50
1 atm
x=0.50
x=0.25
x=0.17

GLASS MANUFACTURING APPARATUS AND GLASS MANUFACTURING METHOD

TECHNICAL FIELD

[0001] The present disclosure relates to a glass manufacturing apparatus and a glass manufacturing method.

BACKGROUND ART

[0002] A glass manufacturing apparatus includes a burner that forms a flame inside a glass melting furnace. The burner burns a flammable gas and a oxidizing gas to form a flame. The flammable gas is, for example, natural gas. The natural gas contains CH_4 gas as the major component. The oxidizing gas is, for example, pure oxygen gas or air. The flame heats a glass raw material and molten glass obtained by melting the glass raw material.

[0003] Air may be used as the oxidizing gas as described above. In this case, because N_2 gas, which makes up the majority of the air, does not contribute to combustion, a large amount of air is used and a large amount of combustion gas is emitted from a glass melting furnace. The combustion gas is a gas that remains after a combustion reaction of the flammable gas and the oxidizing gas (including N_2 gas that does not contribute to a combustion reaction). A first regenerator and a second regenerator may be provided adjacent to the glass melting furnace so that heat is collected from a large amount of combustion gas that has been emitted.

[0004] The first regenerator and the second regenerator are used in combination with first and second burners. A glass manufacturing apparatus alternately and iteratively performs a process in which the first burner forms a flame inside the glass melting furnace and the second burner forms a flame inside the glass melting furnace.

[0005] The first regenerator collects heat from the combustion gas emitted from the glass melting furnace while the second burner forms a flame inside the glass melting furnace. The first regenerator releases the collected heat to heat at least one of the flammable gas and the oxidizing gas while the first burner forms a flame inside the glass melting furnace. The first burner burns at least one of the flammable gas and the oxidizing gas heated in the first regenerator to form a flame inside the glass melting furnace.

[0006] Likewise, the second regenerator collects heat from the combustion gas emitted from the glass melting furnace while the first burner forms a flame inside the glass melting furnace. The second regenerator releases the collected heat to heat at least one of the flammable gas and the oxidizing gas while the second burner forms a flame inside the glass melting furnace. The second burner burns at least one of the flammable gas and the oxidizing gas heated in the second regenerator to form a flame inside the glass melting furnace.

[0007] The first regenerator disclosed in Patent Document 1 releases heat while the first burner forms a flame inside the glass melting furnace, thereby performing an endothermic reaction. As an example of the endothermic reaction, a reaction for producing CO gas and H_2 gas from CH_4 gas, H_2O gas, and CO_2 gas is disclosed. The CO gas and the H_2 gas produced in the first regenerator are supplied to the first burner as the flammable gas.

[0008] Moreover, pure oxygen gas may be used as the oxidizing gas as described above. In this case, the oxidizing gas does not contain N_2 gas. Therefore, because an amount of combustion gas to be emitted is small and the regenerator is not required, a degree of freedom in the design of the glass melting furnace is increased. Moreover, N_2gas, which does not contribute to combustion, is not ineffectively heated and glass can be efficiently heated by combustion heat. Furthermore, the production of NO_x gas can be suppressed.

[0009] A recuperator may be used instead of a regenerator as a means for collecting the heat of the combustion gas emitted from the glass melting furnace.

[0010] The recuperator of Patent Document 2 heats air through heat exchange with the combustion gas. Air transports heat from the recuperator to a reactor. The reactor heats natural gas through heat exchange with air heated in the recuperator to perform an endothermic reaction for producing CH_4 gas from C_mH_n gas contained in the natural gas (m and n are integers of 2 or more).

[0011] Patent Document 3 discloses that when an H_2O gas concentration in the furnace atmosphere of the glass melting furnace is excessively high, a moisture concentration contained in the molten glass is excessively high and the glass quality is low (for example, a defect occurs in a bottom surface of the float glass).

CITATION LIST

Patent Document

[0012] Patent Document 1: Japanese Patent No. 3705713

[0013] Patent Document 2: Japanese Patent No. 6980795

[0014] Patent Document 3: Japanese Patent No. 6144622

SUMMARY OF INVENTION

Technical Problem

[0015] Natural gas is commonly used as a flammable gas that is burned by a burner. The natural gas mainly contains CH_4 gas. When the CH_4 gas is burned, CO_2 gas is produced and the CO_2 gas is emitted. Therefore, a process in which H_2 gas is used instead of the CH_4 gas so that an amount of CO_2 gas to be produced and an amount of CO_2 gas to be emitted are reduced is conceived.

[0016] When the H_2 gas is burned instead of the CH_4 gas, the H_2O gas is produced by a reaction between the H_2 gas and O_2 gas and the CO_2 gas is not produced. However, the H_2O gas concentration in the furnace atmosphere is high because CO_2 gas is not produced. This is noticeable when pure oxygen gas is used as a oxidizing gas. When the pure oxygen gas is used, the H_2O gas concentration in the furnace atmosphere can theoretically be 100% by volume.

[0017] In addition, the glass raw material may emit a gas during melting. For example, when the glass raw material contains a carbonate such as calcium carbonate or magnesium carbonate, the glass raw material releases the CO_2 gas during melting.

[0018] Therefore, the H_2O gas concentration in the furnace atmosphere is actually less than 100% by volume.

[0019] When the H_2O gas concentration in the furnace atmosphere is excessively high, NaOH gas derived from Na contained in the molten glass is easily produced and furnace materials such as refractories are easily corroded. Alternatively, when the H_2O gas concentration in the furnace atmosphere is excessively high, the moisture concentration in the molten glass may be excessively high and the glass quality may be low.

[0020] An aspect of the present disclosure provides a technique for reducing an H_2O gas concentration in a furnace atmosphere of a glass melting furnace when H_2 gas is used as a flammable gas.

Solution to Problem

[0021] According to an aspect of the present disclosure, there is provided a glass manufacturing apparatus including: a recuperator configured to heat a heat medium through heat exchange with a combustion gas emitted from a glass melting furnace; a first reactor configured to produce CH_4 gas and H_2O gas from CO_2 gas and H_2 gas by performing a methanation reaction; an water vapor controller configured to adjust an H_2O concentration in a product of the first reactor by removing at least a part of the H_2O gas from the product of the first reactor; a second reactor configured to produce CO gas and H_2 gas from the product of the first reactor with the H_2O concentration adjusted in the water vapor controller by performing at least one of a dry reforming reaction and a steam reforming reaction through heat exchange with the heat medium heated by the recuperator; and a burner configured to form a flame in the glass melting furnace by burning a oxidizing gas and a flammable gas containing the CO gas and the H_2 gas produced by the second reactor.

Advantageous Effects of Invention

[0022] According to the aspect of the present disclosure, when the H_2 gas is used as the flammable gas, a transformation is performed to produce at least the CO gas from the H_2 gas and the CO_2 gas, and the flammable gas containing the produced CO gas and the oxidizing gas are burned by the burner. Thereby, the H_2O gas concentration in the combustion gas can be reduced and the H_2O gas concentration in a furnace atmosphere of the glass melting furnace can be reduced. Moreover, by removing at least the part of the H_2O gas after the methanation reaction and before at least one of the dry reforming reaction and the steam reforming reaction, the H_2O gas concentration in the furnace atmosphere of the glass melting furnace can be further reduced.

BRIEF DESCRIPTION OF DRAWINGS

[0023] FIG. **1** A diagram showing a glass manufacturing apparatus according to a reference embodiment.

[0024] FIG. **2** A diagram showing an example (x=0.5) of a reaction in which an overall reaction is represented by Eq. (1).

[0025] FIG. **3** A diagram showing Examples 1 to 14 of combinations of a flammable gas, a oxidizing gas, an H_2O gas concentration in a combustion gas, and an H_2O gas concentration in a furnace atmosphere.

[0026] FIG. **4** A diagram showing a first embodiment of a glass manufacturing apparatus including a recuperator and a reactor.

[0027] FIG. **5** A diagram showing a second embodiment of a glass manufacturing apparatus including a recuperator and a reactor.

[0028] FIG. **6** A diagram showing a third embodiment of a glass manufacturing apparatus including a recuperator and a reactor.

[0029] FIG. **7** A diagram showing a fourth embodiment of a glass manufacturing apparatus including a recuperator and a reactor.

[0030] FIG. **8** A diagram showing a fifth embodiment of a glass manufacturing apparatus including a recuperator and a reactor.

[0031] FIG. **9** A diagram showing a sixth embodiment of a glass manufacturing apparatus including a recuperator and a reactor.

[0032] FIG. **10** A diagram showing a first embodiment of a glass manufacturing apparatus including a first regenerator and a second regenerator.

[0033] FIG. **11** A diagram showing a second embodiment of a glass manufacturing apparatus including a first regenerator and a second regenerator.

[0034] FIG. **12** A diagram showing a third embodiment of a glass manufacturing apparatus including a first regenerator and a second regenerator.

[0035] FIG. **13** A diagram showing a fourth embodiment of a glass manufacturing apparatus including a first regenerator and a second regenerator.

[0036] FIG. **14** A diagram showing an example of a condition for suppressing a byproduct of C in the second reactor.

[0037] FIG. **15** A diagram showing an example of an equilibrium composition when a gas mixture of an initial composition shown at point P**1** (a=0.5 and b=0.5) in FIG. **14** is heated.

[0038] FIG. **16** A diagram showing an example of a relationship of a molar ratio x, a CH_4 selectivity, and a temperature in the first reactor.

DESCRIPTION OF EMBODIMENTS

[0039] Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. In each drawing, the same or corresponding constituent elements are denoted by the same reference signs and the description thereof may be omitted. In the description, "to (~)" indicating a numerical range means that numerical values written before and after "to (~)" include a lower limit value and an upper limit value.

[0040] A glass manufacturing apparatus **1** according to a reference embodiment will be described with reference to FIG. **1**. The glass manufacturing apparatus **1** includes a glass melting furnace **10**. The glass melting furnace **10** houses a glass raw material and molten glass obtained by melting the glass raw material. After the molten glass is removed from the glass melting furnace **10**, the molten glass is molded into a desired shape and slowly cooled. Thereby, a glass product is obtained.

[0041] The glass raw material is prepared by mixing several types of materials. The glass raw material may contain a fining agent. The glass raw material may include glass cullet to recycle glass. The glass raw material may be a powdered raw material or a granulated raw material obtained by granulating the powdered raw material. The glass raw material is decided in accordance with a composition of the glass product.

[0042] The glass melting furnace **10** includes refractories. The refractories include, for example, zirconia-based fused cast refractories, alumina-based fused cast refractories, alumina-zirconia-based fused cast refractories, Al—Zr—Si (AZS)-based fused cast refractories, dense bonded refractories, and the like. The glass melting furnace **10** may include a plurality of types of refractory refractories.

[0043] The glass manufacturing apparatus **1** includes a burner **20**. The burner **20** forms a flame inside the glass melting furnace **10**. The flame heats the glass raw material and the molten glass. The glass raw material is input from above with respect to a surface level of the molten glass and forms a layer on at least a part of the surface level. The glass raw material is gradually fused into the molten glass. Although only one of the burners **20** is shown in FIG. **1**, the number of burners **20** is typically two or more.

[0044] In addition, the glass manufacturing apparatus **1** may use a plurality of electrodes (not shown) as a heat source. The plurality of electrodes perform a current-carrying process on the molten glass by applying an alternating current (AC) voltage to the molten glass. In this case, the molten glass is heated. The glass manufacturing apparatus **1** may use an electric heater (not shown) as a heat source. The electric heater itself produces heat. It is only necessary for the glass manufacturing apparatus **1** to include at least the burner **20** as a heat source.

[0045] The burner **20** forms a flame by burning a flammable gas and a oxidizing gas. In the present reference embodiment, at least a part of the flammable gas is H_2 gas (more specifically, a gas obtained by transforming the H_2 gas as will be described below). The use of the H_2 gas can reduce the amount of CO_2 gas that is produced and the amount of CO_2 gas to be emitted compared with the use of CH_4 gas.

[0046] In addition, it is only necessary for the H_2 gas to be used as at least a part of the flammable gas and it is only necessary to use the natural gas in combination therewith. Because the natural gas contains CH_4 gas as a main component, an amount of natural gas to be used is more preferably smaller. The amount of natural gas to be used is preferably 0% to 50% by volume, more preferably 0% to 25% by volume, and even more preferably 0% by volume, from the viewpoint of reducing the amount of CO_2 gas to be produced.

[0047] When the H_2 gas is burned instead of the CH_4 gas, the H_2O gas is produced by a reaction between the H_2 gas and O_2 gas, and the CO_2 gas is not produced. However, an H_2O gas concentration in the furnace atmosphere is high because the CO_2 gas is not produced. This is noticeable when pure oxygen gas is used as the oxidizing gas. When the pure oxygen gas is used, the H_2O gas concentration in the furnace atmosphere can theoretically be 100% by volume.

[0048] In addition, the glass raw material may release a gas during melting. For example, if the glass raw material contains a carbonate such as calcium carbonate or magnesium carbonate, the glass raw material releases CO_2 gas during melting. Therefore, the H_2O gas concentration in the furnace atmosphere is actually less than 100% by volume.

[0049] If the H_2O gas concentration in the furnace atmosphere is excessively high, NaOH gas derived from Na contained in the molten glass is easily produced and furnace materials such as refractories are easily corroded. Alternatively, when the H_2O gas concentration in the furnace atmosphere is excessively high, a moisture concentration in the molten glass may be excessively high and the glass quality may be low.

[0050] If air is used instead of pure oxygen gas as the oxidizing gas, the H_2O gas concentration in the furnace atmosphere can be reduced. In this case, because the N_2gas, which makes up the majority of the air, does not contribute to combustion, a large amount of air is used and a large amount of combustion gas is emitted from the glass melting furnace. The combustion gas is a gas that remains after a combustion reaction of the flammable gas and the oxidizing gas (including the N_2 gas that does not contribute to combustion). To collect the heat of the large amount of combustion gas that has been emitted, a regenerator is provided adjacent to the glass melting furnace **10**.

[0051] On the other hand, if the pure oxygen gas is used as the oxidizing gas, the oxidizing gas does not contain N_2 gas. Therefore, because an amount of combustion gas to be emitted is small and the regenerator is not required, the degree of freedom in the design of the glass melting furnace **10** is increased. Moreover, the N_2 gas, which does not contribute to combustion, is not ineffectively heated and glass can be efficiently heated by combustion heat. Furthermore, the production of NO_x gas can be suppressed.

[0052] However, if the pure oxygen gas is used as the oxidizing gas, an H_2O gas concentration in the furnace atmosphere is higher than when air is used. When oxygen-enriched air is used instead of pure oxygen gas as the oxidizing gas, the H_2O gas concentration is higher than when air is used. The oxygen-enriched air is a gas mixture of pure oxygen gas and air and is a gas having a higher oxygen gas concentration than air.

[0053] The glass manufacturing apparatus **1** of the present reference embodiment includes a gas transformer **30** that transforms the H_2 gas to reduce the H_2O gas concentration in the furnace atmosphere. One or more gas transformers **30** (one gas transformer **30** in FIG. **1**) are provided. The gas transformer **30** produces at least the CO gas and the H_2O gas from the CO_2 gas and the H_2 gas according to a reaction in which an overall reaction is represented by the following Eq. (1) or (2). The CO_2 gas and the H_2 gas are reactants and the CO gas and the H_2O gas are products. The product may further include at least one of the H_2 gas and the CO_2 gas.

[Chem. 1]

$$xCO_2+H_2 \rightarrow xCO+(1-x)H_2+xH_2O\ (0<x\leq 1) \tag{1}$$

[Chem. 2]

$$xCO_2+H_2 \rightarrow (x-1)CO_2+CO+H_2O\ (1<x) \tag{2}$$

[0054] In Eqs. (1) and (2), x denotes a molar ratio of the CO_2 gas to the H_2 gas supplied to the gas transformer **30**. The molar ratio x is greater than 0.00. In general, the molar ratio is consistent with a volume ratio.

[0055] In addition, it is only necessary for the gas transformer **30** to perform a reaction in which the overall reaction is represented by Eq. (1) or Eq. (2) and this reaction may be simultaneously performed with a reaction different therefrom. Thus, it is only necessary to supply the CO_2 gas and the H_2 gas as the reactants to the gas transformer **30**, and other gases may be further supplied.

[0056] In FIG. **2**, an example (x=0.5) of a reaction in which the overall reaction is given by Eq. (1) is shown. In FIG. **2**, ΔH is a variation in enthalpy (KJ/mol). The enthalpy is also referred to as heat content. As can be seen from FIG. **2**, the reaction in which the overall reaction is represented by Eq. (1) or (2) is an endothermic reaction. The product of the endothermic reaction (the CO gas) potentially contains more heat than the reactant of the endothermic reaction (the H_2 gas).

[0057] Reactions in which the overall reaction is represented by Eq. (1) or (2) include, for example, at least one of (A) a reverse shift (a reverse water gas shift: RWGS) reaction, (B) a combination of a methanation reaction and a dry reforming (DRM) reaction, and (C) a combination of a methanation reaction and a steam reforming (STR) reaction.

[0058] Compared to the above-described (B) and (C), the above-described (A) can reduce the number of reactions used and simplify the apparatus configuration. On the other hand, the above-described (B) and (C) can lower the reaction temperature compared to the above-described (A). Incidentally, the reverse shift reaction is represented by the following Eq. (3). The methanation reaction is represented by the following Eq. (4). The DRM reaction is represented by the following Eq. (5). In addition, the STR reaction is represented by the following Eq. (6).

[Chem. 3]

$$CO_2+H_2+41.2[kJ/mol]\rightarrow CO+H_2O \quad (3)$$

[Chem. 4]

$$CO_2+4H_2\rightarrow CH_4+2H_2+165.0[kJ/mol] \quad (4)$$

[Chem. 5]

$$CH_4+CO_2+247.3[kJ/mol]\rightarrow 2CO+2H_2 \quad (5)$$

[Chem. 6]

$$CH_4+H_2O+206.2[kJ/mol]\rightarrow CO+3H_2 \quad (6)$$

[0059] From Eq. (3), it can be seen that the reverse shift reaction is an endothermic reaction. From Eq. (4), it can be seen that the methanation reaction is an exothermic reaction. From Eq. (5), it can be seen that the DRM reaction is the endothermic reaction. From Eq. (6), it can be seen that the STR reaction is the endothermic reaction.

[0060] The product of the endothermic reaction (the CO gas) potentially contains more heat than the reactant of the endothermic reaction (the H_2 gas or the CH_4 gas). Therefore, the combustion of the product of the endothermic reaction (the CO gas) yields more combustion heat than the combustion of the reactant of the endothermic reaction (the H_2 gas or the CH_4 gas).

[0061] The gas transformer **30** supplies a product such as the CO gas to the burner **20**. The gas to be supplied includes, for example, the CO gas and the H_2O gas (see Eqs. (1) and (2)). The gas to be supplied may further include at least one of the H_2 gas and the CO_2 gas.

[0062] The burner **20** forms a flame inside the glass melting furnace **10** by burning a flammable gas containing the CO gas produced by the gas transformer **30** and a oxidizing gas. The combustion of the CO gas improves the heat of combustion per mole more than the combustion of the H_2 gas.

[0063] The combustion reaction of the CO gas is represented by the following Eq. (7) and the combustion reaction of the H_2 gas is represented by the following Eq. (8). Here, a lower heating value is used for the heat of combustion of the H_2 gas.

[Chem. 7]

$$CO+\frac{1}{2}O_2\rightarrow CO_2+282.2\ [kJ/mol] \quad (7)$$

-continued

[Chem. 8]

$$H_2+\frac{1}{2}O_2\rightarrow H_2O+241.9\ [kJ/mol] \quad (8)$$

[0064] From Eqs. (7) and (8), it can be seen that combustion of the CO gas can improve the heat of combustion per mole more than the combustion of the H_2 gas.

[0065] The combustion reaction of the flammable gas and the oxidizing gas by the burner **20** is represented by, for example, the following Eq. (9) or (10).

[Chem. 9]

$$xCO+(1-x)H_2+xH_2O+\frac{1}{2}O_2\rightarrow xCO_2+H_2O\ (0<x\leq 1) \quad (9)$$

[Chem. 10]

$$(x-1)CO_2+CO+H_2O+\frac{1}{2}O_2\rightarrow xCO_2+H_2O\ (1<x) \quad (10)$$

[0066] Eq. (9) represents the combustion reaction of the product obtained in Eq. (1). On the other hand, Eq. (10) shows the combustion reaction of the product obtained in Eq. (2). In Eqs. (9) and (10), x denotes a molar ratio of the CO_2 gas to the H_2 gas supplied to the gas transformer **30**. The molar ratio x is greater than 0.00.

[0067] In addition, the combustion reaction of the flammable gas and the oxidizing gas may include at least the combustion reaction represented by Eqs. (9) and (10) and may further include another combustion reaction.

[0068] As described above, the burner **20** burns the flammable gas containing the CO gas produced by the gas transformer **30** and the oxidizing gas. Therefore, the combustion gas includes the CO_2 gas as well as the H_2O gas. This is also obvious from Eq. (9) and Eq. (10). The H_2O gas can be diluted with the CO_2 gas and the H_2O gas concentration in the furnace atmosphere can be reduced.

[0069] Examples 1 to 14 of combinations of the flammable gas, the oxidizing gas, the H_2O gas concentration in the combustion gas, and the H_2O gas concentration in the furnace atmosphere are shown in FIG. **3**. The furnace atmosphere includes a gas released by the glass raw material when the glass raw material is fused, in addition to the combustion gas. Therefore, the H_2O gas concentration in the furnace atmosphere is lower than the H_2O gas concentration in the combustion gas. The H_2O gas concentration in the combustion gas shown in FIG. **3** is a numerical value of a case where the gas transformer **30** does not have the water vapor controller **73** to be described below. When the gas transformer **30** has the water vapor controller **73**, the H_2O gas concentration in the combustion gas is less than the numerical value shown in FIG. **3**.

[0070] As shown in FIG. **3**, if the H_2 gas is used instead of the CH_4 gas as the flammable gas, the H_2O gas concentration in the furnace atmosphere is high because the CO_2 gas is not produced. This is noticeable when pure oxygen gas is used as the oxidizing gas.

[0071] In addition, the H_2O gas concentration in the furnace atmosphere depends on a type of glass raw material, but is mainly determined in accordance with a type of flammable gas and a type of oxidizing gas. Therefore, in the present reference embodiment, the H_2O gas concentration in the combustion gas is adopted as a parameter to be managed.

[0072] The H_2O gas concentration in the combustion gas is preferably 67% or less by volume. Molten glass has been conventionally manufactured by burning the CH_4 gas and the O_2 gas (pure oxygen gas). It has been confirmed that if the H_2O gas concentration in the combustion gas is 67% or less by volume, there is no problem with the quality of the glass product or corrosion of the furnace material.

[0073] Preferably, the molar ratio x is 0.50 or more so that the H_2O gas concentration in the combustion gas is 67% or less by volume (see FIG. **3**). The H_2O gas concentration in the combustion gas is more preferably 65% or less by volume. More preferably, the molar ratio x is 0.55 or more so that the H_2O gas concentration in the combustion gas is 65% or less by volume (see FIG. **3**).

[0074] Although a lower limit of the H_2O gas concentration in the combustion gas is not particularly limited, the concentration is preferably 20% or more by volume from the viewpoint of feasibility. Preferably, the molar ratio x is 4.00 or less so that the H_2O gas concentration in the combustion gas is 20% or more by volume (see FIG. **3**).

[0075] More preferably, the molar ratio x is less than or equal to 1.00. If the molar ratio x is 1.00 or less, the combustion reaction is represented by Eq. (9). On the other hand, when the molar ratio x exceeds 1.00, the combustion reaction is represented by Eq. (10). As can be seen from Eq. (10), there is CO_2 gas of (x-1) moles which does not contribute to combustion when the molar ratio x exceeds 1.00. This CO_2 gas not only does not produce combustion heat but also consumes combustion heat ineffectively. If the molar ratio x is 1.00 or less, CO_2 gas of (x-1) moles which does not contribute to combustion will not be ineffectively heated and the glass will be heated efficiently with combustion heat.

[0076] Although a dedicated supply source such as a gas cylinder (not shown) may be used to supply the CO_2 gas to the gas transformer **30**, a circulation line **50** is used in the present reference embodiment. The circulation line **50** supplies at least a part of the CO_2 gas contained in the combustion gas from the glass melting furnace **10** to the gas transformer **30**. At least a part of the CO_2 gas emitted from the glass melting furnace **10** can be used as a reactant in the overall reaction. The CO_2 gas can be circulated and the amount of CO_2 gas to be emitted can be reduced.

[0077] A first dehydration unit **51** may be provided along the circulation line **50**. The first dehydration unit **51** separates the H_2O gas and the CO_2 gas from each other by removing the H_2O gas from the combustion gas. At least a part of the separated CO_2 gas can be used as a reactant in the overall reaction. The H_2O gas supply to the gas transformer **30** can be limited, and the H_2O gas concentration in the furnace atmosphere of the glass melting furnace **10** can be reduced. The first dehydration unit **51** may remove at least a part of the H_2O gas, but substantially all of the H_2O gas is removed in the present reference embodiment.

[0078] The glass manufacturing apparatus **1** preferably collects heat from the combustion gas burned by the burner **20** and uses the collected heat to perform the reaction in which the overall reaction is represented by Eq. (1) or (2). By collecting heat from the burned combustion gas, the waste of energy can be reduced. As a heat collection means, a recuperator or a regenerator to be described below is used. Hereinafter, the heat collection means and the like will be described.

[0079] A first embodiment of the glass manufacturing apparatus **1** including a recuperator **60** will be described with reference to FIG. **4**. The glass manufacturing apparatus **1** includes a glass melting furnace **10**, a burner **20**, the recuperator **60**, a first reactor **71**, and a second reactor **72**. The gas transformer **30** includes the first reactor **71**, the second reactor **72**, and the like (including an water vapor controller **73** and an additive line **75** to be described below). The first reactor **71** performs the methanation reaction and the second reactor **72** performs at least one of the DRM and STR reactions. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **1** will be mainly described.

[0080] The recuperator **60** heats a heat medium through heat exchange with a combustion gas emitted from the glass melting furnace **10**. Although the heat medium may be a gas or a liquid, the heat medium is preferably a gas from the viewpoint of preventing boiling because the temperature of the combustion gas is high. The heat medium is preferably air. The recuperator **60** transfers heat between the combustion gases and the heat medium (e.g., air) without mixing them. The recuperator **60** is provided outside the glass melting furnace **10** in FIG. **4**, but may be provided inside the glass melting furnace **10**. One or more recuperators **60** (one recuperator **60** in FIG. **4**) are provided. The recuperator **60** is thermally connected to the second reactor **72** via a first heat transport line **80**. The first heat transport line **80** transports the heat medium from the recuperator **60** to the second reactor **72**.

[0081] The first reactor **71** produces CH_4 and H_2O gases from CO_2 and H_2 gases by performing a methanation reaction. The CO_2 and H_2 gases are reactants and the CH_4 and H_2O gases are products (see Eq. (4)). The methanation reaction is sufficiently performed even at a low temperature of 400° C. or lower.

[0082] The second reactor **72** produces CO gas from the product of the first reactor **71** by performing at least one of the DRM and STR reactions through heat exchange with a heat medium heated in the recuperator **60**. When both the DRM and STR reactions occur, the CH_4 and H_2O gases and the CO_2 gas are reactants and the CO and H_2 gases are products (see Eqs. (5) and (6)). The product may include at least one of gases such as the H_2O gas, the CO_2 gas, and the CH_4 gas other than the CO gas and the H_2 gas. The second reactor **72** does not mix the reactant and product of the second reactor **72** with a heat medium (e.g., air) and transfers heat therebetween.

[0083] Because the DRM reaction and the STR reaction are endothermic reactions, heat can be efficiently collected from the heat medium and heat collected from the combustion gas can be used efficiently. The product of the endothermic reaction (the CO gas) also potentially contains more heat than the reactant of the endothermic reaction (the H_2 gas). Therefore, the combustion of the CO gas can improve the heat of combustion per mole more than the combustion of the H_2 gas (see Eqs. (**7**) and (**8**)).

[0084] It is only necessary for the reaction occurring in the second reactor **72** to be an endothermic reaction as a whole. The reaction occurring in the second reactor **72** may include a reaction other than the DRM reaction and the STR reaction. Therefore, in the second reactor **72**, the reactant and product are not particularly limited. The product produced in the second reactor **72** (i.e., the flammable gas) is supplied to the burner **20** via the first supply line **21**. On the other hand, the oxidizing gas is supplied to the burner **20** via the second supply line **22**.

[0085] The glass manufacturing apparatus **1** includes an water vapor controller **73**. The water vapor controller **73** is provided along a connection line **74** connecting the first reactor **71** and the second reactor **72**. The water vapor controller **73** adjusts an H_2O concentration in the product of the first reactor **71** by removing at least a part of the H_2O gas from the product of the first reactor **71**. The water vapor controller **73** is removed, for example, by cooling and liquefying the H_2O gas. The second reactor **72** produces the CO gas from the product of the first reactor **71** in which the H_2O concentration is adjusted in the water vapor controller **73**.

[0086] In the present embodiment, the water vapor controller **73** removes at least a part of the H_2O gas from the product of the first reactor **71** and delivers the product of the first reactor **71** to the second reactor **72**. The H_2O concentration in the product of the second reactor **72** can be reduced and the H_2O concentration in the furnace atmosphere of the glass melting furnace **10** can be reduced.

[0087] A process of removing at least a part of the H_2O gas from the product of the first reactor **71** and delivering the product of the first reactor **71** to the second reactor **72** is also effective in performing the overall reaction represented by Eq. (1) or (2). This is evident from the fact that H_2O does not appear as the substance (reactant) on the left side of the general reaction equation, but is present as the substance (product) on the right side of the general reaction equation. By performing the above-described overall reaction and the endothermic reaction, the heat collection rate can be improved.

[0088] Preferably, the amount of H_2O gas removed in the water vapor controller **73** is set so that C is not produced as a byproduct in the second reactor **72** (or so that soot is not deposited). The second reactor **72** performs at least one of the DRM reaction (see Eq. (5)) and the STR reaction (see Eq. (6)) as described above to produce the CO and H_2gases from at least one of the H_2O and CO_2 gases and the CH_4 gas.

[0089] As can be seen from Eqs. (5) and (6), if the total number of moles of the H_2O and CO_2 gases is less than the number of moles of the CH_4 gas, the CH_4 gas remains. If the CH_4 gas remains, because the reaction temperature in the second reactor **72** is 700° C. or higher, C will be produced as a byproduct according to pyrolysis of the CH_4 gas. Thus, it is preferable to set the amount of H_2O gas removal in the water vapor controller **73** so that soot is not deposited.

[0090] Next, the amount of H_2O gas removed in the water vapor controller **73** will be described with reference to FIGS. **14** and **15**. In FIGS. **14** and **15**, a denotes a molar ratio (CO_2/CH_4) of the CO_2 gas supplied to the second reactor **72** to the CH_4 gas supplied to the second reactor **72** and b denotes a molar ratio (H_2O/CH_4) of the H_2O gas supplied to the second reactor **72** to the CH_4 gas supplied to the second reactor **72**.

[0091] In order to prevent the byproduct of C in the second reactor **72**, the water vapor controller **73** preferably removes only a part of the H_2O gas from the product of the first reactor **71** so that the following Inequality (a1) is valid when a reaction temperature in the second reactor **72** is 900° C. or higher (preferably 1600° C. or lower), the following Inequality (a2) is valid when the reaction temperature is 800° C. or higher and lower than 900° C., and the following Inequality (a3) is valid when the reaction temperature is 700° C. or higher and lower than 800° C.

$$b \geq -a + 1 \tag{a1}$$

$$b \geq -0.68a + 1.01 \tag{a2}$$

$$b \geq -0.1143a^2 + 0.0286a + 1.3029 \tag{a3}$$

[0092] Here, the condition for preventing the byproduction of C in the second reactor **72** was obtained by calculating the equilibrium composition at atmospheric pressure at each temperature with a gas mixture in which CH_4 gas of 1 mole, CO_2 gas of a moles, and H_2O gas of b moles are mixed as the initial composition. The equilibrium composition was calculated using the thermodynamic equilibrium calculation software (a software name of FactSage developed by GTT-Technologies).

[0093] In FIG. **15**, an example of an equilibrium composition when a gas mixture of an initial composition shown at point **P1** (a=0.5 and b=0.5) in FIG. **14** is heated is shown. As can be seen from FIG. **15**, when the initial composition is (a=0.5 and b=0.5), the byproduct of C can be prevented if the temperature is 900° C. or higher. As can be seen from FIG. **15**, if a and b are constant, the byproduct of C is more easily suppressed when the temperature is higher.

[0094] The inventors of the present invention have derived the above Inequalities (a1), (a2), and (a3) by calculating the minimum temperature required to prevent the byproduct of C using a and b as variables. In addition, when Inequality (a1) is valid, it indicates that the sum of a and b is 1 or more and indicates that the total number of moles of the H_2O and CO_2 gases is greater than or equal to the number of moles of the CH_4 gas. When the above Inequality (a1) is not valid, the total number of moles of the H_2O gas and the CO_2 gas is less than the number of moles of the CH_4 gas, and the excess of the CH_4 gas results in the byproduct of C even when the reaction temperature is high.

[0095] More preferably, the water vapor controller **73** removes only a part of the H_2O gas from the product of the first reactor **71** so that the following Inequality (a4) is valid in addition to any one of the above Inequalities (a1), (a2), and (a3).

$$b \geq 2a \tag{a4}$$

[0096] If the above Inequality (a4) is valid, an H_2O gas concentration in the combustion gas of the burner **20** is 67% or less by volume. It has been confirmed that if the H_2O gas concentration in the combustion gas is 67% or less by volume, there is no problem with the quality of the glass product or corrosion of the furnace material.

[0097] Moreover, more preferably, the water vapor controller **73** removes only a part of the H_2O gas from the product of the first reactor **71** so that the following Inequality (a5) is valid in addition to any one of the above Inequalities (a1), (a2), and (a3).

$$b \geq -a + 3 \tag{a5}$$

[0098] If the above Inequality (a5) is valid, a coking reaction in the surface of the metal catalyst can be suppressed when the second reactor **72** has the metal catalyst. The coking reaction is a reaction in which soot is deposited in a kinetically favorable side reaction, even under equilibrium conditions in which soot is not deposited. If soot covers the surface of the metal catalyst, the activity of the metal catalyst is reduced.

[0099] In addition, when the second reactor **72** does not have a metal catalyst, the above Inequality (a5) may or may not be valid. When the second reactor **72** does not have a metal catalyst, the following Inequality (a6) may be valid instead of the above Inequality (a5).

$$b<-a+3. \tag{a6}$$

[0100] If the above Inequality (a6) is valid, the amount of H_2O gas removal in the water vapor controller **73** is large and the H_2O gas concentration in the combustion gas is low.

[0101] The glass manufacturing apparatus **1** preferably includes an additive line **75**. The additive line **75** supplies at least a part of the CO_2 gas for use in the DRM reaction to the second reactor **72** without passing through the first reactor **71**. Thereby, it is possible to reduce a molar ratio x in the first reactor **71** and suppress the deposition of soot in the first reactor **71**.

[0102] Next, an example of a relationship of the molar ratio x, the CH_4 selectivity, and the temperature in the first reactor **71** will be described with reference to FIG. **16**. The molar ratio x is a molar ratio of the CO_2 gas to the H_2 gas as described above. The CH_4 selectivity is a ratio of an amount of CH_4 gas production to a total amount of product production in a methanation reaction (a reaction in which the CH_4 gas and the H_2O gas are produced from the CO_2 gas and the H_2 gas).

[0103] As shown in FIG. **16**, the CH_4 selectivity is low if the temperature exceeds 500° C. When the temperature exceeds 500° C., the CH_4 gas is expected to decompose. Accordingly, the temperature of the first reactor **71** is preferably 500° C. or lower, and more preferably 400° C. or lower. The methanation reaction is sufficiently performed even at a temperature of 400° C. or lower.

[0104] Moreover, as shown in FIG. **16**, when the molar ratio x is greater than 0.25, the CH_4 rselectivity is low. It is considered that if the molar ratio x exceeds 0.25, carbon is precipitated (or soot is deposited). Accordingly, it is preferable to have a molar ratio x of 0.25 or less in at least the first reactor **71** so that the deposition of soot in the first reactor **71** is suppressed.

[0105] In addition, the molar ratio x in the gas transformer **30**, which includes the first reactor **71**, the second reactor **72**, and the like, may exceed 0.25. It is only necessary for the molar ratio x in the first reactor **71** to be 0.25 or less.

[0106] CO_2 gas is used in both the methanation reaction and the DRM reaction. Thus, although it is necessary to supply the CO_2 gas to the first reactor **71**, it is not necessary to supply the CO_2 gas for use in both the first reactor **71** and the second reactor **72** to the first reactor **71**.

[0107] The additive line **75** supplies at least a part of the CO_2 gas for use in the second reactor **72** to the second reactor **72** without passing through the first reactor **71** so that the deposition of soot in the first reactor **71** is suppressed as described above. Thereby, it is possible to reduce the molar ratio x in the first reactor **71** and suppress the deposition of soot in the first reactor **71**.

[0108] The additive line **75** branches from the circulation line **50**, which will be discussed below, for example, to bypass the first reactor **71** and join the connection line **74**. The additive line **75** may bypass not only the first reactor **71** but also the water vapor controller **73**. Cooling of the CO_2 gas can be suppressed by the water vapor controller **73**. A flow rate controller **76** may be provided along the additive line **75**.

[0109] Although the additive line **75** joins a connection line **74**, the additive line **75** may be connected to the second reactor **72** without joining the connection line **74**. In the former case, the product of the first reactor **71** and the CO_2 gas are premixed and supplied to the second reactor **72**. In the latter case, the product of the first reactor **71** and the CO_2 gas are separately supplied to the second reactor **72**.

[0110] The glass manufacturing apparatus **1** preferably includes a circulation line **50**. The circulation line **50** supplies at least a part of the CO_2 gas contained in the combustion gas burned by the burner **20** from the glass melting furnace **10** to the first reactor **71**. At least a part of the CO_2 gas emitted from the glass melting furnace **10** can be used as a reactant in the methanation reaction. The CO_2 gas can be circulated and an amount of CO_2 gas to be emitted can be reduced. At least a part of the CO_2 gas emitted from the glass melting furnace **10** can also be used as a reactant in the DRM or STR reaction.

[0111] The glass manufacturing apparatus **1** may include a first dehydration unit **51** along the circulation line **50**. The first dehydration unit **51** separates the H_2O gas from the CO_2 gas by removing the H_2O gas from the combustion gas. It is possible to restrict the circulation of the H_2O gas while permitting the circulation of CO_2 gas. The H_2O gas not only does not produce heat of combustion but also ineffectively consumes heat of combustion. If the circulation of H_2O gas can be restricted, the non-flammable H_2O gas can be avoided, and the heat of combustion can be used to heat glass efficiently. If the circulation of H_2O gas can be restricted, the concentration of H_2O gas in the furnace atmosphere can be reduced.

[0112] Although it is only necessary for the first dehydration unit **51** to remove at least a part of the H_2O gas, substantially all of the H_2O gas is removed in the present embodiment. When only a part of the H_2O gas is removed from the first dehydration unit **51** or when there is no first dehydration unit **51** as described below, the gas containing the H_2O gas is supplied to the first reactor **71**.

[0113] More preferably, a desulfurization unit **52** is provided along the circulation line **50**. The desulfurization unit **52** removes a sulfur-containing gas from the combustion gas. The sulfur contained in the sulfur-containing gas is derived from a glass raw material, for example, a fining agent. The supply of the sulfur-containing gas to the first reactor **71** can be restricted by providing the desulfurization unit **52** along the circulation line **50**. Thereby, the first reactor **71** can be maintained in a clean state.

[0114] Because the second reactor **72** uses a heat medium to collect heat from the combustion gas unlike a regenerator to be described below, it is possible to restrict a process in which the combustion gas flows into the second reactor **72** as it is and it is possible to maintain the second reactor **72** in a clean state. Therefore, when the second reactor **72** has a metal catalyst, the deterioration of the metal catalyst can be suppressed.

[0115] The second reactor **72** may have a metal catalyst. The metal catalyst promotes at least one of the DRM reaction and the STR reaction. The metal catalyst can increase a reaction rate of at least one of the DRM reaction and the STR reaction and cause the reaction to equilibrate in a short time. The metal catalyst may also reduce a volume of the second reactor **72** by increasing the reaction rate. As a metal catalyst, copper (Cu), nickel (Ni), or the like is used. If the reaction rate is sufficiently fast, the metal catalyst may be eliminated.

[0116] A second embodiment of a glass manufacturing apparatus **1** including a recuperator **60** will be described with reference to FIG. **5**. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **4** will be mainly described. As shown in FIG. **5**, the glass manufacturing apparatus **1** may not include the first dehydration unit **51** (see FIG. **4**). An water vapor controller **73** may also serve as the first dehydration unit **51**.

[0117] A third embodiment of a glass manufacturing apparatus **1** including a recuperator **60** will be described with reference to FIG. **6**. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **4** will be mainly described. As shown in FIG. **6**, the water vapor controller **73** may include a dehydration unit **73***a* in which a part of H_2O gas is removed by cooling and liquefying and a bypass line **73***b* along which the remaining part of the H_2O gas is delivered from a first reactor **71** to a second reactor **72**.

[0118] According to the present embodiment, all unnecessary H_2O gas can be removed in the dehydration unit **73***a* while the H_2O gas for use in the STR reaction remains. Moreover, the H_2O gas for use in the STR reaction can also be delivered from the first reactor **71** to the second reactor **72** without cooling in the dehydration unit **73***a*. The bypass line **73***b* bypasses the dehydration unit **73***a*. A flow rate controller **73***c* may be provided along the bypass line **73***b*.

[0119] A fourth embodiment of a glass manufacturing apparatus **1** including a recuperator **60** will be described with reference to FIG. **7**. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **4** will be mainly described. As shown in FIG. **7**, the glass manufacturing apparatus **1** may include a first heat exchanger **77** and a second heat transport line **78**. The first heat exchanger **77** is provided along a connection line **74** and is thermally connected to a first reactor **71** via a second heat transport line **78**.

[0120] The first reactor **71** heats a heat medium (e.g., air) using heat produced in a methanation reaction. The first reactor **71** does not mix the reactant or product of the methanation reaction with the heat medium and transfers heat therebetween. The second heat transport line **78** transports the heat medium heated in the first reactor **71** to the first heat exchanger **77**.

[0121] The first heat exchanger **77** heats the product of the first reactor **71** with the H_2O concentration adjusted in the water vapor controller **73** using the heat produced in the methanation reaction before the product is supplied to the second reactor **72**. The heat produced by the methanation reaction can be effectively used to reduce the specific energy consumption of molten glass. The specific energy consumption is a consumption amount of energy input from outside the system to produce 1 ton of molten glass. The first heat exchanger **77** does not mix the product of the first reactor **71** with the heat medium and transfers heat therebetween.

[0122] In addition, the heat produced by the methanation reaction may be used in a different field. For example, the first heat exchanger **77** may be provided along the second supply line **22** to preheat the oxidizing gas (e.g., pure oxygen gas). Heat produced in the methanation reaction may not be used inside the system or may be used outside the system.

[0123] A fifth embodiment of a glass manufacturing apparatus **1** including a recuperator **60** will be described with reference to FIG. **8**. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **4** will be mainly described. As shown in FIG. **8**, the glass manufacturing apparatus **1** preferably includes a second heat exchanger **23**.

[0124] The second heat exchanger **23** is provided along the second supply line **22** and preheats a oxidizing gas (e.g., pure oxygen gas) through heat exchange with a heat medium heated by the recuperator **60**. Heat collected from a combustion gas can be effectively used to reduce the specific energy consumption of molten glass. The second heat exchanger **23** does not mix the oxidizing gas with the heat medium (e.g., air) and transfers heat therebetween.

[0125] The first heat transport line **80** transports the heat medium from the recuperator **60** to the second heat exchanger **23**. Preferably, the first heat transport line **80** transports the heat medium from the recuperator **60** to the second reactor **72** and then transports the heat medium from the second reactor **72** to the second heat exchanger **23**. Before the oxidizing gas is heated, the reactant of the second reactor **72** can be heated to easily secure heat necessary to perform a reaction in the second reactor **72**.

[0126] A sixth embodiment of a glass manufacturing apparatus **1** including a recuperator **60** will be described with reference to FIG. **9**. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **4** will be mainly described. As shown in FIG. **9**, the glass manufacturing apparatus **1** preferably includes a third heat exchanger **24** and a second dehydration unit **25**.

[0127] The third heat exchanger **24** transfers heat from the product of the second reactor **72** to the oxidizing gas at a first point of the first supply line **21** overlapping the second supply line **22**. The third heat exchanger **24** does not mix the product of the second reactor **72** with the oxidizing gas and transfers heat therebetween. The second dehydration unit **25** removes the H_2O gas from the product of the second reactor **72** at a second point located downstream from the first point of the first supply line **21**. Although it is only necessary for the second dehydration unit **25** to remove at least a part of the H_2O gas, substantially all of the H_2O gas is removed in the present embodiment.

[0128] The second dehydration unit **25** removes the H_2O gas from the product of the second reactor **72**, for example, by cooling the product of the second reactor **72**. The H_2O gas not only does not produce heat of combustion but also ineffectively consumes heat of combustion. The second dehydration unit **25** limits the supply of the H_2O gas to the burner **20**. It is possible to efficiently heat glass with combustion heat without ineffectively heating H_2O gas not contributing to combustion. Moreover, the H_2O gas concentration in the furnace atmosphere can be reduced.

[0129] Before the second dehydration unit **25** cools the product of the second reactor **72**, the third heat exchanger **24** transfers heat from the product of the second reactor **72** to the oxidizing gas. Thereby, heat collected from the combustion gas can be used efficiently, thereby reducing the specific energy consumption of the molten glass.

[0130] A first embodiment of a glass manufacturing apparatus **1** including a first regenerator **90**A and a second regenerator **90**B will be described with reference to FIG. **10**. The glass manufacturing apparatus **1** includes a glass melting furnace **10**, a first burner **20**A, a second burner **20**B, the first regenerator **90**A, the second regenerator **90**B, and a first reactor **71**. A gas transformer **30** includes the first regenerator **90**A, the second regenerator **90**B, the first reactor **71**, or the like (including an water vapor controller **73** and an additive line **75** to be described below). The first reactor **71** performs a methanation reaction and the first regenerator **90**A and the second regenerator **90**B perform at least one of DRM and STR reactions. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **1** will be mainly described.

[0131] The first regenerator **90**A and the second regenerator **90**B are used in combination with the first burner **20**A and the second burner **20**B. Although the number of first regenerators **90**A is one in FIG. **10**, the number of first regenerators **90**A may be two or more. Likewise, it is only necessary for the number of second regenerators **90**B, the number of first burners **20**A, and the number of second burners **20**B to be one or more.

[0132] The glass manufacturing apparatus **1** alternately and iteratively performs a process in which the first burner **20**A forms a flame inside the glass melting furnace **10** and the second burner **20**B forms a flame inside the glass melting furnace **10**. FIG. **10** is a diagram showing a state in which the first burner **20**A forms a flame inside the glass melting furnace **10**.

[0133] The first regenerator **90**A collects heat from a combustion gas emitted from the glass melting furnace **10** while the second burner **20**B forms a flame inside the glass melting furnace **10**. The first regenerator **90**A produces CO gas from the product of the first reactor **71** by releasing the heat collected in advance and performing at least one of the DRM and STR reactions while the first burner **20**A forms a flame inside the glass melting furnace **10**. The first burner **20**A burns a flammable gas containing the CO gas produced by the first regenerator **90**A and a oxidizing gas to form a flame inside the glass melting furnace **10**.

[0134] Likewise, the second regenerator **90**B collects heat from the combustion gas emitted from the glass melting furnace **10** while the first burner **20**A forms a flame inside the glass melting furnace **10**. The second regenerator **90**B produces the CO gas from the product of the first reactor **71** by releasing the heat collected in advance and performing at least one of the DRM and STR reactions while the second burner **20**B forms a flame inside the glass melting furnace **10**. The second burner **20**B burns the flammable gas containing the CO gas produced by the second regenerator **90**B and the oxidizing gas to form a flame inside the glass melting furnace **10**.

[0135] Unlike the recuperator **60** and the first reactor **71**, the first regenerator **90**A and the second regenerator **90**B direct the flow of combustion gas. The heat of the combustion gas is accumulated in the structures of the first regenerator **90**A and the second regenerator **90**B.

[0136] The first regenerator **90**A and the second regenerator **90**B produce CO gas from the product of the first reactor **71** by performing at least one of the DRM and STR reactions. When both the DRM and STR reactions occur, CH_4 and H_2O gases and CO_2gas are reactants and CO and H_2 gases are products (see Eqs. (5) and (6)). The product may include a gas other than the CO gas and the H_2 gas, for example, at least one of H_2O gas, CO_2 gas, and CH_4 gas.

[0137] Because the DRM reaction and the STR reaction are endothermic reactions, heat can be efficiently collected from the heat medium and heat collected from the combustion gas can be used efficiently. Moreover, the product of the endothermic reaction (the CO gas) potentially contains more heat than the reactant of the endothermic reaction (the H_2 gas). Therefore, the combustion of the CO gas can improve the heat of combustion per mole (see Eqs. (7) and (8)) as compared to the combustion of the H_2 gas.

[0138] It is only necessary for the reactions occurring in the first regenerator **90**A and the second regenerator **90**B to be endothermic reactions as a whole. The reactions occurring in the first regenerator **90**A and the second regenerator **90**B may include reactions other than DRM reactions and STR reactions. Therefore, in the first regenerator **90**A and the second regenerator **90**B, the reactants and products are not particularly limited.

[0139] The glass manufacturing apparatus **1** includes an water vapor controller **73**. The water vapor controller **73** is provided along a connection line **74** connecting the first reactor **71** and the first regenerator **90**A. The water vapor controller **73** adjusts an H_2O concentration in the product of the first reactor **71** by removing at least a part of the H_2O gas from the product of the first reactor **71**. The water vapor controller **73** is removed, for example, by cooling and liquefying the H_2O gas. The first regenerator **90**A produces CO gas from the product of the first reactor **71** in which the H_2O concentration is adjusted by the water vapor controller **73**.

[0140] In the present embodiment, the water vapor controller **73** removes at least a part of the H_2O gas from the product of the first reactor **71** and then delivers the product of the first reactor **71** to the first regenerator **90**A. The H_2O concentration in the product of the first regenerator **90**A can be reduced and the H_2O concentration in the furnace atmosphere of the glass melting furnace **10** can be reduced.

[0141] A process of removing at least a part of the H_2O gas from the product of the first reactor **71** and then delivering the product of the first reactor **71** to the first regenerator **90**A is also effective in performing the overall reaction represented by Eq. (1) or (2). This is evident from the fact that H_2O does not appear as the substance (reactant) on the left side of the general reaction equation, but is present as the substance (product) on the right side of the general reaction equation. By performing the above-described overall reaction and endothermic reaction, the heat collection rate can be improved.

[0142] Although the connection line **74** may remove at least a part of the H_2O gas from the product of the first reactor **71** in the water vapor controller **73** and then deliver the product of the first reactor **71** to the first regenerator **90**A, the product may be delivered to the second regenerator **90**B instead of the first regenerator **90**A. In addition, the connection line **74** connecting the first reactor **71** and the first regenerator **90**A and the connection line **74** connecting the first reactor **71** and the second regenerator **90**B may be different or may be provided separately.

[0143] Preferably, the amount of H_2O gas removal in the water vapor controller **73** is set so that C is not produced as a byproduct in the first regenerator **90**A (or so that soot is not deposited). Specifically, the water vapor controller **73** preferably removes only a part of the H_2O gas from the product of the first reactor **71** so that the following Inequality (b1) is valid when a reaction temperature in the first regenerator **90**A is 900° C. or higher, the following Inequality (b2) is valid when the reaction temperature is 800° C. or higher and lower than 900° C., and the following Inequality (b3) is valid when the reaction temperature is 700° C. or higher and lower than 800° C. while the first burner **20**A forms the flame inside the glass melting furnace **10**.

$$b \geq -a + 1 \tag{b1}$$

$$b \geq -0.68a + 1.01 \tag{b2}$$

$$b \geq -0.1143a^2 + 0.0286a + 1.3029 \tag{b3}$$

[0144] In the above Inequalities (b1), (b2), and (b3), a denotes a molar ratio (CO_2/CH_4) of the CO_2 gas supplied to the first regenerator **90**A to the CH_4 gas supplied to the first regenerator **90**A and b denotes a molar ratio (H_2O/CH_4) of the H_2O gas supplied to the first regenerator **90**A to the CH_4 gas supplied to the first regenerator **90**A.

[0145] More preferably, the following Inequality (b4) is valid in addition to any one of the above-described Inequalities (b1), (b2), and (b3).

$$b \leq 2a \tag{b4}$$

[0146] If the above Inequality (b4) is valid, an H_2O gas concentration in the combustion gas of the first burner **20**A is 67% or less by volume. It has been confirmed that if the H_2O gas concentration in the combustion gas is 67% or less by volume, there is no problem with the quality of the glass product or corrosion of the furnace material.

[0147] In addition to any one of the above Inequalities (b1), (b2), and (b3), the following Inequality (b5) may be valid.

$$b \geq -a + 3 \tag{b5}$$

[0148] The first regenerator **90**A does not have a metal catalyst. Therefore, the following Inequality (b6) may be valid instead of the above-described Inequality (b5).

$$b < -a + 3 \tag{b6}$$

[0149] If the above Inequality (b6) is valid, the amount of H_2O gas removal in the water vapor controller **73** is high and the H_2O gas concentration in the combustion gas is low.

[0150] Likewise, preferably, the amount of H_2O gas removal in the water vapor controller **73** is set so that C is not produced as a byproduct in the second regenerator **90**B (or so that soot is not deposited). Specifically, the water vapor controller **73** removes only a part of the H_2O gas from the product of the first reactor **71** so that the following Inequality (c1) is valid when a reaction temperature in the second regenerator **90**B is 900° C. or higher (preferably 1600° C. or lower), the following Inequality (c2) is valid when the reaction temperature is 800° C. or higher and lower than 900° C., and the following Inequality (c3) is valid when the reaction temperature is 700° C. or higher and lower than 800° C. while the second burner **20**B forms the flame inside the glass melting furnace **10**.

$$b \geq -a + 1 \tag{c1}$$

$$b \geq -0.68a + 1.01 \tag{c2}$$

$$b \geq -0.1143a^2 + 0.0286a + 1.3029 \tag{c3}$$

[0151] In the above Inequalities (c1), (c2), and (c3), a denotes a molar ratio (CO_2/CH_4) of the CO_2 gas supplied to the second regenerator **90**B to the CH_4 gas supplied to second regenerator **90**B and b denotes a molar ratio (H_2O/CH_4) of the H_2O gas supplied to the second regenerator **90**B to the CH_4 gas supplied to the second regenerator **90**B.

[0152] More preferably, the following Inequality (c4) is valid in addition to any one of the above-described Inequalities (c1), (c2), and (c3).

$$b \leq 2a \tag{c4}$$

[0153] If the above Inequality (c4) is valid, an H_2O gas concentration in the combustion gas of the second burner **20**B is 67% or less by volume. It has been confirmed that if the H_2O gas concentration in the combustion gas is 67% or less by volume, there is no problem with the quality of the glass product or corrosion of the furnace material.

[0154] In addition to any one of the above Inequalities (c1), (c2), and (c3), the following Inequality (c5) may be valid.

$$b \geq -a + 3 \tag{c5}$$

[0155] The second regenerator **90**B does not have a metal catalyst. Therefore, the following Inequality (c6) may be valid instead of the above-described Inequality (c5).

$$b < -a + 3 \tag{c6}$$

[0156] If the above Inequality (c6) is valid, the amount of H_2O gas removal in the water vapor controller **73** is high and the H_2O gas concentration in the combustion gas is low.

[0157] The glass manufacturing apparatus **1** preferably includes an additive line **75**. The additive line **75** supplies at least a part of the CO_2 gas for use in the DRM reaction to the first regenerator **90**A without passing through the first reactor **71**. Thereby, it is possible to reduce the molar ratio x in the first reactor **71** and suppress the deposition of soot in the first reactor **71**.

[0158] The additive line **75** branches from a circulation line **50**, which will be discussed below, for example, to bypass the first reactor **71** and join the connection line **74**. The additive line **75** may bypass not only the first reactor **71** but also the water vapor controller **73**. Cooling of the CO_2

gas can be suppressed by the water vapor controller **73**. A flow rate controller **76** may be provided along the additive line **75**.

[0159] Although the additive line **75** joins a connection line **74**, the additive line **75** may be connected to the first regenerator **90**A without joining the connection line **74**. In the former case, the product of the first reactor **71** and the CO_2 gas are premixed and supplied to the first regenerator **90**A. In the latter case, the product of the first reactor **71** and the CO_2 gas are separately supplied to the first regenerator **90**A.

[0160] Although the additive line **75** supplies at least a part of the CO_2 gas for use in the DRM reaction to the first regenerator **90**A without passing through the first reactor **71**, it may be supplied to the second regenerator **90**B instead of the first regenerator **90**A. In addition, the additive line **75** for supplying the CO_2 gas to the first regenerator **90**A and the additive line **75** for supplying the CO_2 gas to the second regenerator **90**B may be different from each other or may be provided separately.

[0161] The glass manufacturing apparatus **1** preferably includes a circulation line **50**. The circulation line **50** supplies at least a part of the CO_2 gas contained in the combustion gas emitted from the glass melting furnace **10** from the second regenerator **90**B to the first reactor **71**. At least the part of the CO_2 gas emitted from the glass melting furnace **10** can be used as a reactant in the methanation reaction. The CO_2 gas can be circulated and an amount of CO_2 gas to be emitted can be reduced. At least the part of the CO_2gas emitted from the glass melting furnace **10** can also be used as a reactant in the DRM or STR reaction.

[0162] The circulation line **50** may supply at least the part of the CO_2 gas contained in the combustion gas emitted from the glass melting furnace **10** from the first regenerator **90**A to the first reactor **71**. In addition, the circulation line **50** for delivering the CO_2 gas from the second regenerator **90**B to the first reactor **71** and the circulation line **50** for delivering the CO_2 gas from the first regenerator **90**A to the first reactor **71** may be different or may be provided separately.

[0163] The glass manufacturing apparatus **1** may include a first dehydration unit **51** along the circulation line **50**. The first dehydration unit **51** separates the H_2O gas from the CO_2 gas by removing the H_2O gas from the combustion gas. It is possible to restrict the circulation of the H_2O gas while permitting the circulation of CO_2 gas. The H_2O gas not only does not produce heat of combustion but also ineffectively consumes heat of combustion. If the circulation of the H_2O gas can be restricted, it is possible to efficiently heat glass with combustion heat without ineffectively heating H_2O gas not contributing to combustion. Moreover, if the circulation of the H_2O gas can be restricted, the H_2O gas concentration in the furnace atmosphere can be reduced.

[0164] Although it is only necessary for the first dehydration unit **51** to remove at least a part of the H_2O gas, substantially all of the H_2O gas is removed in the present embodiment. In addition, when only a part of the H_2O gas is removed from the first dehydration unit **51** or when there is no first dehydration unit **51** as described below, the gas containing the H_2O gas is supplied to the first reactor **71**.

[0165] More preferably, a desulfurization unit **52** is provided along the circulation line **50**. The desulfurization unit **52** removes a sulfur-containing gas from the combustion gas. The sulfur contained in the sulfur-containing gas is derived from a glass raw material, for example, a fining agent. The supply of the sulfur-containing gas to the first reactor **71** can be restricted by providing the desulfurization unit **52** along the circulation line **50**.

[0166] A second embodiment of a glass manufacturing apparatus **1** including a first regenerator **90**A and a second regenerator **90**B will be described with reference to FIG. **11**. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **10** will be mainly described. As shown in FIG. **11**, the glass manufacturing apparatus **1** may not include the first dehydration unit **51** (see FIG. **10**). The water vapor controller **73** may also serve as the first dehydration unit **51**.

[0167] A third embodiment of a glass manufacturing apparatus **1** including a first regenerator **90**A and a second regenerator **90**B will be described with reference to FIG. **12**. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **10** will be mainly described. As shown in FIG. **12**, the water vapor controller **73** may include a dehydration unit **73***a* for removing a part of the H_2O gas by cooling and liquefying and a bypass line **73***b* for delivering the remaining part of the H_2O gas from the first reactor **71** to the first regenerator **90**A (or the second regenerator **90**B).

[0168] According to the present embodiment, all unnecessary H_2O gas can be removed in the dehydration unit **73***a* while the H_2O gas for use in the STR reaction remains. Moreover, the H_2O gas for use in the STR reaction can also be delivered from the first reactor **71** to the first regenerator **90**A (or the second regenerator **90**B) without cooling in the dehydration unit **73***a*. The bypass line **73***b* bypasses the dehydration unit **73***a*. A flow rate controller **73***c* may be provided along the bypass line **73***b*.

[0169] A fourth embodiment of a glass manufacturing apparatus **1** including a first regenerator **90**A and a second regenerator **90**B will be described with reference to FIG. **13**. Hereinafter, differences from the glass manufacturing apparatus **1** shown in FIG. **10** will be mainly described. As shown in FIG. **13**, the glass manufacturing apparatus **1** may include a first heat exchanger **77** and a second heat transport line **78**. The first heat exchanger **77** is provided along a connection line **74** and is thermally connected to a first reactor **71** via a second heat transport line **78**.

[0170] The first reactor **71** heats a heat medium (e.g., air) using heat produced in a methanation reaction. The first reactor **71** does not mix the reactant or product of the methanation reaction with the heat medium and transfers heat therebetween. The second heat transport line **78** transports the heat medium heated in the first reactor **71** to the first heat exchanger **77**.

[0171] The first heat exchanger **77** heats the product of the first reactor **71** with the H_2O concentration adjusted in the water vapor controller **73** using the heat produced in the methanation reaction before the product is supplied to the first regenerator **90**A (or the second regenerator **90**B). The heat produced by the methanation reaction can be effectively used to reduce the specific energy consumption of molten glass. The first heat exchanger **77** does not mix the product of the first reactor **71** with the heat medium and transfers heat therebetween.

[0172] In addition, the heat produced by the methanation reaction may be used in a different field. For example, the first heat exchanger **77** may be provided along the second supply line **22** to preheat the oxidizing gas (e.g., pure oxygen gas). Heat produced in the methanation reaction may not be used inside the system or may be used outside the system.

[0173] The following appendixes are disclosed with respect to the above-described embodiments and the like.

[Appendix 1] A glass manufacturing apparatus including:

[0174] a recuperator configured to heat a heat medium through heat exchange with a combustion gas emitted from a glass melting furnace;

[0175] a first reactor configured to produce CH_4 gas and H_2O gas from CO_2 gas and H_2 gas by performing a methanation reaction;

[0176] an water vapor controller configured to adjust an H_2O concentration in a product of the first reactor by removing at least a part of the H_2O gas from the product of the first reactor;

[0177] a second reactor configured to produce CO gas and H_2 gas from the product of the first reactor with the H_2O concentration adjusted in the water vapor controller by performing at least one of a dry reforming reaction and a steam reforming reaction through heat exchange with the heat medium heated by the recuperator; and

[0178] a burner configured to form a flame in the glass melting furnace by burning a oxidizing gas and a flammable gas containing the CO gas and the H_2 gas produced by the second reactor.

[Appendix 2] The glass manufacturing apparatus according to appendix 1,

[0179] wherein the water vapor controller removes only a part of the H_2O gas from the product of the first reactor so that the following Inequality (a1) is valid when a reaction temperature in the second reactor is 900° C. or higher, the following Inequality (a2) is valid when the reaction temperature is 800° C. or higher and lower than 900° C., and the following Inequality (a3) is valid when the reaction temperature is 700° C. or higher and lower than 800° C., and

[0180] wherein, in the following Inequalities (a1), (a2), and (a3), a denotes a molar ratio (CO_2/CH_4) of the CO_2 gas supplied to the second reactor to the CH_4 gas supplied to the second reactor and b denotes a molar ratio (H_2O/CH_4) of the H_2O gas supplied to the second reactor to the CH_4 gas supplied to the second reactor.

$$b \geq -a + 1 \tag{a1}$$

$$b \geq -0.68a + 1.01 \tag{a2}$$

$$b \geq -0.1143a^2 + 0.0286a + 1.3029 \tag{a3}$$

[Appendix 3] The glass manufacturing apparatus according to appendix 2, wherein the water vapor controller removes only a part of the H_2O gas from the product of the first reactor so that the following Inequality (a4) is valid in addition to any one of the above Inequalities (a1), (a2), and (a3).

$$b \leq 2a \tag{a4}$$

[Appendix 4] The glass manufacturing apparatus according to any one of appendixes 1 to 3, including an additive line along which at least a part of the CO_2 gas for use in the dry reforming reaction is supplied to the second reactor without passing through the first reactor.

[Appendix 5] The glass manufacturing apparatus according to any one of appendixes 1 to 4, including a first heat exchanger configured to heat the product of the first reactor with the H_2O concentration adjusted in the water vapor controller using heat produced by the methanation reaction before the product of the first reactor is supplied to the second reactor.

[Appendix 6] The glass manufacturing apparatus according to any one of appendixes 1 to 5, wherein the oxidizing gas is pure oxygen gas or oxygen-enriched air.

[Appendix 7] The glass manufacturing apparatus according to any one of appendixes 1 to 6, including a circulation line along which at least a part of the CO_2 gas contained in a combustion gas burned by the burner is supplied from the glass melting furnace to the first reactor.

[Appendix 8] The glass manufacturing apparatus according to appendix 7, including a first dehydration unit configured to remove the H_2O gas from the combustion gas along the circulation line.

[Appendix 9] The glass manufacturing apparatus according to any one of appendixes 1 to 8, including a second heat exchanger configured to preheat the oxidizing gas through the heat exchange with the heat medium heated by the recuperator.

[Appendix 10] The glass manufacturing apparatus according to any one of appendixes 1 to 9, including:

[0181] a first supply line along which a product of the second reactor is supplied to the burner;

[0182] a second supply line along which the oxidizing gas is supplied to the burner;

[0183] a third heat exchanger configured to transfer heat from the product of the second reactor to the oxidizing gas at a first point of the first supply line overlapping the second supply line; and

[0184] a second dehydration unit configured to remove the H_2O gas from the product of the second reactor at a second point located downstream from the first point of the first supply line.

[Appendix 11] The glass manufacturing apparatus according to any one of appendixes 1 to 10, wherein the second reactor has a metal catalyst that promotes at least one of the dry reforming reaction and the steam reforming reaction.

[Appendix 12] A glass manufacturing apparatus for alternately and iteratively performing a process in which a first burner forms a flame inside a glass melting furnace and a second burner forms a flame inside the glass melting furnace, the glass manufacturing apparatus including:

[0185] a first reactor configured to produce CH_4 gas and H_2O gas from CO_2 gas and H_2 gas by performing a methanation reaction;

[0186] an water vapor controller configured to adjust an H_2O concentration in a product of the first reactor by removing at least a part of the H_2O gas from the product of the first reactor;

[0187] a first regenerator configured to produce CO gas and the H_2 gas from the product of the first reactor with the H_2O concentration adjusted in the water vapor controller by collecting heat from a combustion gas emitted from the glass melting furnace while the second burner forms the flame inside the glass melting furnace, releasing the heat while the first burner forms the flame inside the glass melting furnace, and performing at least one of a dry reforming reaction and a steam reforming reaction;

[0188] the first burner configured to form the flame inside the glass melting furnace by burning a flammable gas containing the CO gas and the H_2 gas produced in the first regenerator and a oxidizing gas;

[0189] a second regenerator configured to produce the CO gas and the H_2 gas from the product of the first reactor with the H_2O concentration adjusted in the water vapor controller by collecting heat from a combustion gas emitted from the glass melting furnace while the first burner forms the flame inside the glass melting furnace, releasing the heat while the second burner forms the flame inside the glass melting furnace, and performing at least one of the dry reforming reaction and the steam reforming reaction; and

[0190] the second burner configured to form the flame inside the glass melting furnace by burning the flammable gas containing the CO gas and the H_2 gas produced in the second regenerator and the oxidizing gas.

[Appendix 13] The glass manufacturing apparatus according to appendix 12,

[0191] wherein the water vapor controller removes only a part of the H_2O gas from the product of the first reactor so that the following Inequality (b1) is valid when a reaction temperature in the first regenerator is 900° C. or higher, the following Inequality (b2) is valid when the reaction temperature is 800° C. or higher and lower than 900° C., and the following Inequality (b3) is valid when the reaction temperature is 700° C. or higher and lower than 800° C. while the first burner forms the flame inside the glass melting furnace, and

[0192] wherein, in the following Inequalities (b1), (b2), and (b3), a denotes a molar ratio (CO_2/CH_4) of the CO_2 gas supplied to the first regenerator to the CH_4 gas supplied to the first regenerator and b denotes a molar ratio (H_2O/CH_4) of the H_2O gas supplied to the first regenerator to the CH_4 gas supplied to the first regenerator.

$$b \geq -a + 1 \tag{b1}$$

$$b \geq -0.68a + 1.01 \tag{b2}$$

$$b \geq -0.1143a^2 + 0.0286a + 1.3029 \tag{b3}$$

[Appendix 14] The glass manufacturing apparatus according to appendix 13, wherein the water vapor controller removes only a part of the H_2O gas from the product of the first reactor so that the following Inequality (b4) is valid in addition to any one of the above Inequalities (b1), (b2), and (b3).

$$b \leq 2a \tag{b4}$$

[Appendix 15] The glass manufacturing apparatus according to any one of appendixes 12 to 14, including an additive line along which at least a part of the CO_2 gas for use in the dry reforming reaction is supplied to the first regenerator or the second regenerator without passing through the first reactor.

[Appendix 16] The glass manufacturing apparatus according to any one of appendixes 12 to 15, including a first heat exchanger configured to heat the product of the first reactor with the H_2O concentration adjusted in the water vapor controller using heat produced by the methanation reaction before the product of the first reactor is supplied to the first regenerator or the second regenerator.

[Appendix 17] The glass manufacturing apparatus according to any one of appendixes 12 to 16, wherein the oxidizing gas is pure oxygen gas or oxygen-enriched air.

[Appendix 18] The glass manufacturing apparatus according to any one of appendixes 12 to 17, including a circulation line along which at least a part of the CO_2 gas contained in a combustion gas emitted from the glass melting furnace is supplied from the second regenerator or the first regenerator to the first reactor.

[Appendix 19] The glass manufacturing apparatus according to appendix 18, including a dehydration unit configured to remove the H_2O gas from the combustion gas along the circulation line.

[Appendix 20] A glass manufacturing method including manufacturing molten glass using the glass manufacturing apparatus according to any one of appendixes 1 to 19.

[0193] Although the glass manufacturing apparatus and the glass manufacturing method according to the present disclosure have been described above, the present disclosure is not limited to the above-described embodiments and the like. Various changes, modifications, substitutions, additions, deletions, and combinations can be made without departing from the scope described in the claims. These are, of course, also within the technical scope of the present disclosure.

[0194] Priority is claimed on Japanese Patent Application No. 2022-188270, filed Nov. 25, 2022, the content of which is incorporated herein by reference.

REFERENCE SIGNS LIST

[0195] **1** Glass manufacturing apparatus
[0196] **10** Glass melting furnace
[0197] **20** Burner
[0198] **60** Recuperator
[0199] **71** First reactor
[0200] **72** Second reactor
[0201] **73** Water vapor controller

What is claimed is:

1. A glass manufacturing apparatus comprising:

a recuperator configured to heat a heat medium through heat exchange with a combustion gas emitted from a glass melting furnace;

a first reactor configured to produce CH_4 gas and H_2O gas from CO_2 gas and H_2 gas by performing a methanation reaction;

an water vapor controller configured to adjust an H_2O concentration in a product of the first reactor by removing at least a part of the H_2O gas from the product of the first reactor;

a second reactor configured to produce CO gas and H_2 gas from the product of the first reactor with the H_2O concentration adjusted in the water vapor controller by performing at least one of a dry reforming reaction and a steam reforming reaction through heat exchange with the heat medium heated by the recuperator; and a burner configured to form a flame in the glass melting furnace by burning a oxidizing gas and a flammable gas containing the CO gas and the H_2 gas produced by the second reactor.

2. The glass manufacturing apparatus according to claim **1**,

wherein the water vapor controller removes only a part of the H_2O gas from the product of the first reactor so that the following Inequality (a1) is valid when a reaction temperature in the second reactor is 900° C. or higher, the following Inequality (a2) is valid when the reaction temperature is 800° C. or higher and lower than 900° C., and the following Inequality (a3) is valid when the reaction temperature is 700° C. or higher and lower than 800° C., and

wherein, in the following Inequalities (a1), (a2), and (a3), a denotes a molar ratio (CO_2/CH_4) of the CO_2 gas supplied to the second reactor to the CH_4 gas supplied to the second reactor and b denotes a molar ratio (H_2O/CH_4) of the H_2O gas supplied to the second reactor to the CH_4 gas supplied to the second reactor.

$$b \geq -a+1 \tag{a1}$$

$$b \geq -0.68a+1.01 \tag{a2}$$

$$b \geq -0.1143a^2+0.0286a+1.3029. \tag{a3}$$

3. The glass manufacturing apparatus according to claim **2**, wherein the water vapor controller removes only a part of the H_2O gas from the product of the first reactor so that the following Inequality (a4) is valid in addition to any one of the above Inequalities (a1), (a2), and (a3).

$$b \leq 2a. \tag{a4}$$

4. The glass manufacturing apparatus according to any claim **1**, comprising an additive line along which at least a part of the CO_2 gas for use in the dry reforming reaction is supplied to the second reactor without passing through the first reactor.

5. The glass manufacturing apparatus according to claim **1**, comprising a first heat exchanger configured to heat the product of the first reactor with the H_2O concentration adjusted in the water vapor controller using heat produced by the methanation reaction before the product of the first reactor is supplied to the second reactor.

6. The glass manufacturing apparatus according to claim **1**, wherein the oxidizing gas is pure oxygen gas or oxygen-enriched air.

7. The glass manufacturing apparatus according to claim **1**, comprising a circulation line along which at least a part of the CO_2 gas contained in a combustion gas burned by the burner is supplied from the glass melting furnace to the first reactor.

8. The glass manufacturing apparatus according to claim **7**, comprising a first dehydration unit configured to remove the H_2O gas from the combustion gas along the circulation line.

9. The glass manufacturing apparatus according to claim **1**, comprising a second heat exchanger configured to preheat the oxidizing gas through the heat exchange with the heat medium heated by the recuperator.

10. The glass manufacturing apparatus according to claim **1**, comprising:

a first supply line along which a product of the second reactor is supplied to the burner;

a second supply line along which the oxidizing gas is supplied to the burner;

a third heat exchanger configured to transfer heat from the product of the second reactor to the oxidizing gas at a first point of the first supply line overlapping the second supply line; and

a second dehydration unit configured to remove the H_2O gas from the product of the second reactor at a second point located downstream from the first point of the first supply line.

11. The glass manufacturing apparatus according to any claim **1**, wherein the second reactor has a metal catalyst that promotes at least one of the dry reforming reaction and the steam reforming reaction.

12. A glass manufacturing apparatus for alternately and iteratively performing a process in which a first burner forms a flame inside a glass melting furnace and a second burner forms a flame inside the glass melting furnace, the glass manufacturing apparatus comprising:

a first reactor configured to produce CH_4 gas and H_2O gas from CO_2 gas and H_2 gas by performing a methanation reaction;

an water vapor controller configured to adjust an H_2O concentration in a product of the first reactor by removing at least a part of the H_2O gas from the product of the first reactor;

a first regenerator configured to produce CO gas and the H_2 gas from the product of the first reactor with the H_2O concentration adjusted in the water vapor controller by collecting heat from a combustion gas emitted from the glass melting furnace while the second burner forms the flame inside the glass melting furnace, releasing the heat while the first burner forms the flame inside the glass melting furnace, and performing at least one of a dry reforming reaction and a steam reforming reaction;

the first burner configured to form the flame inside the glass melting furnace by burning a flammable gas containing the CO gas and the H_2 gas produced in the first regenerator and a oxidizing gas;

a second regenerator configured to produce the CO gas and the H_2 gas from the product of the first reactor with the H_2O concentration adjusted in the water vapor controller by collecting heat from a combustion gas emitted from the glass melting furnace while the first burner forms the flame inside the glass melting furnace, releasing the heat while the second burner forms the flame inside the glass melting furnace, and performing at least one of the dry reforming reaction and the steam reforming reaction; and

the second burner configured to form the flame inside the glass melting furnace by burning the flammable gas containing the CO gas and the H_2 gas produced in the second regenerator and the oxidizing gas.

13. The glass manufacturing apparatus according to claim **12**,

wherein the water vapor controller removes only a part of the H_2O gas from the product of the first reactor so that the following Inequality (b1) is valid when a reaction temperature in the first regenerator is 900° C. or higher, the following Inequality (b2) is valid when the reaction temperature is 800° C. or higher and lower than 900° C., and the following Inequality (b3) is valid when the reaction temperature is 700° C. or higher and lower than 800° C. while the first burner forms the flame inside the glass melting furnace, and

wherein, in the following Inequalities (b1), (b2), and (b3), a denotes a molar ratio (CO_2/CH_4) of the CO_2 gas supplied to the first regenerator to the CH_4 gas supplied to the first regenerator and b denotes a molar ratio (H_2O/CH_4) of the H_2O gas supplied to the first regenerator to the CH_4 gas supplied to the first regenerator.

$$b \geq -a+1 \tag{b1}$$

$$b \geq -0.68a+1.01 \tag{b2}$$

$$b \geq -0.1143a^2+0.0286a+1.3029. \tag{b3}$$

14. The glass manufacturing apparatus according to claim **13**, wherein the water vapor controller removes only a part of the H_2O gas from the product of the first reactor so that the following Inequality (b4) is valid in addition to any one of the above Inequalities (b1), (b2), and (b3).

$$b \leq 2a. \tag{b4}$$

15. The glass manufacturing apparatus according to claim **12**, comprising an additive line along which at least a part of the CO_2 gas for use in the dry reforming reaction is supplied to the first regenerator or the second regenerator without passing through the first reactor.

16. The glass manufacturing apparatus according to claim **12**, comprising a first heat exchanger configured to heat the product of the first reactor with the H_2O concentration adjusted in the water vapor controller using heat produced by the methanation reaction before the product of the first reactor is supplied to the first regenerator or the second regenerator.

17. The glass manufacturing apparatus according to claim **12**, wherein the oxidizing gas is pure oxygen gas or oxygen-enriched air.

18. The glass manufacturing apparatus according to claim **12**, comprising a circulation line along which at least a part of the CO_2 gas contained in a combustion gas emitted from the glass melting furnace is supplied from the second regenerator or the first regenerator to the first reactor.

19. The glass manufacturing apparatus according to claim **18**, comprising a dehydration unit configured to remove the H_2O gas from the combustion gas along the circulation line.

20. A glass manufacturing method comprising manufacturing molten glass using the glass manufacturing apparatus according to claim **1**.

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